

Proposal for a Homeland Defense Interceptor



Tactical Interceptor for Targeting and Affordable Neutralization



Submitted by The
University of Texas at
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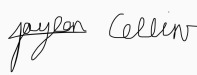
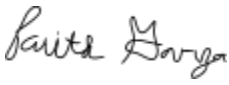
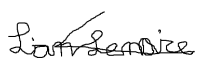
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Nomenclature and Acronyms

AB	After Burner	INEWS	Integrated Electronic Warfare System
APU	Auxiliary Power Unit	IRST	Infrared Search and Track System
CAD	Computer Aided Design	MMH/FH	Maintenance Man Hours to Flight Hours
CAP	Combat Air Patrol	MITR	Maximum Instantaneous Turn Rate
CE	Chief Engineer	NACA	National Advisory Committee for Aeronautics
CG	Center of Gravity	NATO	North Atlantic Treaty Organization
COL	Center of Lift	NMi	Nautical Mile
DAPCA-IV	Development and Procurement Costs of Aircraft	RDT&E	Research, Development, Test, and Evaluation
DCA	Defensive Counter-Air	RFP	Request for Proposal
DGW	Max Design Gross Weight	SEP	Specific Excess Power
ECM	Electronic Countermeasure	SLF	Sustained Load Factor
ECS	Environmental Control System	SM	Static Margin
FEA	Finite Element Analysis	SPGR	Single-Point Ground Refueling
FH	Flight Hours	TOGW	Takeoff gross weights
GFE	Government Furnished Equipment	TRIM	T-38 Talon Repair, Inspection, and Maintenance
GTOW	Gross Takeoff Weight	TSFC	Thrust Specific Fuel Consumption
HDI	Homeland Defense Interceptor	UAV	Unmanned Aerial Vehicle
ICNIA	Integrated Communication, Navigation, and Identification Avionics	USD	United States Dollar
IMR	Intercept Mission Radius		
TITAN	Tactical Interceptor for Targeting and Affordable Neutralization		



1. Proposal Objectives

This proposal has been prepared in response to the Request for Proposal (RFP) for the “Homeland Defense Interceptor,” issued as part of the AIAA Undergraduate Design Team Project for the 2024–2025 academic year. The design was developed by Team TITAN, a student team from the University of Texas at Austin, as part of an academic exercise in advanced aerospace vehicle design. All contributors to this project are undergraduate students and active members of the AIAA student chapter at UT Austin, working collaboratively to meet the performance, cost, and integration requirements outlined in the RFP.

2. Executive Summary

The Homeland Defense Interceptor (HDI) is a high-performance, cost-effective solution developed in response to the United States’ need for rapid-response air defense capabilities. Designed to defend against a diverse array of aerial threats, the HDI emphasizes affordability, simplicity, and mission adaptability. The aircraft features a streamlined, single-engine configuration optimized for supersonic interception, extended loitering, and escort duties. Its aerodynamic airframe, integrated avionics, and internal systems are tailored to meet all mission performance requirements while remaining within the \$25 million flyaway cost cap established in the RFP. Leveraging design inspiration from operationally proven platforms, the HDI balances performance, reliability, and maintainability to ensure consistent operational capability across all mission profiles with minimal logistical burden.

With the increasing emphasis on national security and the need for rapid-response aerial defense systems, the development of an advanced Homeland Defense Interceptor has become a critical objective. Current high-speed interceptor aircraft are either cost-prohibitive or designed for multi-role operations, leading to inefficiencies in dedicated homeland defense applications. The HDI program aims to deliver a cost-effective, highly maneuverable, and supersonic-capable aircraft tailored for defense against modern aerial threats.

This proposal outlines the design of the tactical interceptor for targeting and affordable neutralization aircraft developed by team TITAN. This HDI is designed to provide an optimal balance between speed, endurance, and affordability, ensuring a clear operational advantage. By leveraging a high-thrust, fuel-efficient propulsion system and integrating advanced avionics with a streamlined weapons suite, the HDI will enhance national air defense capabilities while maintaining economic feasibility. The key design objectives focus on achieving superior



thrust-to-weight ratios and ensuring operational flexibility. The major design metrics are outlined in the subsequent sections.

Table 2.1: Weight Summary

Parameter	Value
Gross Takeoff Weight	11574 lbm
Fuel Weight-usable (internal)	2534 lbm
Fuel Weight-trapped	152.07 lbm

Table 2.2: Mission Range Performance

Parameter	Requirement	DCA Mission	Intercept Mission	Escort Mission
Mission Range (nmi)	N/A	800	400	500
Loiter Time (min)	DCA: 240 Intercept: 0 Escort: 60	240	0	60

Table 2.3: Cost Estimates

Parameter	Requirement	Estimate
Unit Flyaway Cost (2024 USD)	< \$25 million	\$16.758 million
Lifecycle Cost (2024 USD)	N/A	\$23.391 million

3. Requirements Compliance

The AIAA RFP outlines mandatory and tradable requirements, in addition to key design objectives. The compliance checklist outlined in the following table includes the aircraft characteristics and compliance of the Homeland Defense Interceptor relevant to these major requirements. Additionally, the aircraft is designed in compliance with relevant military and aviation standards, with detailed justifications provided throughout the report in relevant sections.

Table 3.1: Mission Requirements



Mission Name	Type
Defensive Counter-Air Patrol Mission	Design Mission
Point Defense Intercept Mission	Evaluation Mission
Intercept/ Escort Mission	Evaluation Mission

Table 3.2: Performance Requirements

Criteria	Requirement
Intercept Mission Radius	200 nm
DCA Mission CAP endurance at 300 nm radius	4hrs
Maximum Mach Number at 35,000 ft	Mach 1.6
1-g SEP-Military Thrust at 0.9M/Sea Level	200 ft/sec
1-g SEP-Military Thrust at 0.9M/15,000 ft	50 ft/sec
1-g SEP-Maximum Thrust at 0.9M/Sea Level	700 ft/sec
1-g SEP-Maximum Thrust at 0.9M/15,000 ft	400 ft/sec
5-g SEP-Maximum Thrust at 0.9M/Sea Level	300 ft/sec
5-g SEP-Maximum Thrust at 0.9M/15,000 ft	50 ft/sec
SLF- Maximum Thrust at 0.9M/15,000 ft	5.0 g's
MITR at 35,000 ft	18.0 deg/s

Table 3.3: Weapons Carriage Requirements

Requirement
Trade number of AIM-120 AMRAAM missiles
Trade internal and external carriage

Table 3.4: Engine Cycle Requirements

Requirement
Adequate installed thrust for all phases of the mission
Provide 50 kW of electrical power for aircraft systems
Allocate 2% of engine mass flow to ECS and avionics cooling



The engine must be a non-developmental item in production or expected to be by 2027

Table 3.5: Government Furnished Equipment Requirements

Avionics	ICNIA	Data bus
Flight and Propulsion Control System	Vehicle Management System	-
ECM Equipment	INEWS	-
Fire Control System	IRST System	Active Array Radar
Systems and Equipment	Electrical System	Auxiliary Power Unit
Air-to-Air Weapons	AIM-120 AMRAAM	M61A1 20 mm Cannon

Table 3.6: Other Capabilities and Characteristics Requirements

RFP Requirement	Specification
Remote Pilot	All systems must be designed for remote pilot operation.
Maintenance	The design must allow easy access to and removal of the primary elements of all major systems
Structural Load Factor	+7/-3 g (50% internal fuel)
Safety Factor	1.5
Structural Design Max Dynamic Pressure	2,133 psf (M=1.2 at sea level)
Design Service life	2,000 hours
Fuel/Fuel Tanks	Primary design fuel is standard JP-8 or Jet-A (6.67 lb/gal). All tanks should be self-sealing.
Stability	Unaugmented subsonic longitudinal static margin (S.M.) shall be no greater than 10% and no less than -10%.
Operation	The aircraft must operate in all weather from existing NATO runways (8,000 ft), shelters, and maintenance facilities, and from austere bases without support equipment.
Max Flyaway Cost	Flyaway cost per aircraft for a 1000-aircraft buy will not exceed \$25 million in 2024 US dollars.

All performance analysis was performed using Raymer's Equations.



4. Concept of Operations

4.1 Mission Adaptability and Performance Requirements

The Homeland Defense Interceptor (HDI) is designed to fulfill three primary mission profiles as outlined in the RFP: Defensive Counter-Air Patrol, Point Defense Interception, and Escort Operations. Each of these missions imposes distinct demands on the interceptor's performance envelope, including specific requirements for range, endurance, climb rate, maneuverability, and weapon deployment.

In a DCA mission, the aircraft must loiter at high altitudes for extended periods, maintaining readiness while minimizing fuel consumption. For point defense, it must rapidly climb and dash at supersonic speeds to intercept incoming threats with precision and speed. Escort operations demand coordinated flight with other assets, requiring a stable aerodynamic platform and reliable systems under varied conditions.

These mission profiles necessitate a carefully optimized balance between fuel efficiency, propulsion power, and combat capability. The HDI must consistently deliver high-speed dashes, extended loitering capability, and agile maneuvering while maintaining low operational and lifecycle costs. Ensuring dependable performance across all mission sets is central to achieving the operational goals of the Homeland Defense Interceptor program.

4.2 Missions Overview

4.2.1 Defensive Counter-Air Patrol Mission

The Defensive Counter-Air (DCA) Patrol Mission is focused on maintaining continuous airspace control over a designated area, ensuring a rapid response to potential threats. The mission requires the aircraft to take off, climb to an optimal cruise altitude, and patrol a 300 nm radius while conserving fuel for an extended loiter period. Once on station, the aircraft must conduct four hours of combat air patrol (CAP) at 35,000 feet, remaining ready to engage enemy aircraft or incoming threats.

To effectively conduct the DCA mission, the HDI must feature efficient fuel consumption, advanced sensors for threat detection, and a reliable communications network for coordination with command and control assets. In a high-threat environment, the aircraft must be capable of executing high-speed dashes of up to 100 nm at maximum speed, enabling rapid repositioning for engagement. Combat maneuverability is a critical requirement, with fuel reserves allocated for sustained 360° turns at Mach 1.2 and Mach 0.9 before weapon deployment. After



neutralizing threats, the HDI must return to base via a 400 nm cruise at optimal speed and altitude, descending to sea level before landing. The mission concludes with a 30-minute fuel reserve at sea level to ensure operational safety.

Table 4.2.1: Defensive Counter-Air Patrol Mission Phases

Segment	Description
0-1	Takeoff from land runway (45-60 degrees)
1-2	Climb at maximum thrust (9250 lbf) until 45,000 feet
2-3	Cruise to patrol zone for 300 nm, 45,000 feet, and around Mach 0.8
3-4	Loiter air patrol for 4 hours at 35000 feet and Mach 0.6, ensure continuous surveillance via array radar.
4-5	Engagement, dash 100 nm at maximum speed and 35000 feet. Execute engagement maneuvers and use afterburners if necessary. Use 2 rounds of missiles (2 missiles per round)
5-6	Exit patrol area, climb to optimum speed and altitude, cruise back 400nm, and descend to sea level without fuel usage.
6-7	Land back on the runway, post-flight instructions, and debriefing.

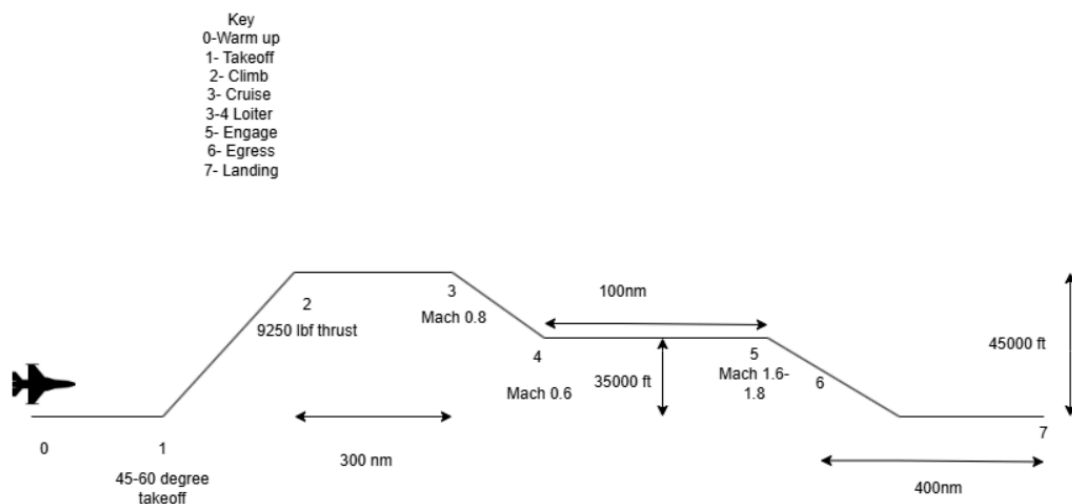


Figure 4.2.1: DCA Patrol Mission Profile

4.2.2 Point Defense Intercept Mission



The Point Defense Interception Mission is centered on rapid response to imminent airborne threats, requiring the HDI to achieve high rates of acceleration, climb, and maneuverability. This mission's profile is depicted in Figure 4.2.2 below, and Table 4.2.2 describes each segment of the mission.

Table 4.2.2: Point Defense Intercept Mission Phases

Segment	Description
1-2	Take-off and Acceleration to Climb Speed at Sea Level
2-3	Climb to 35,000 ft and Accelerate to Max Speed
3-4	Dash 200 nm at Max Speed
4-5	Combat Maneuvers
5-6	Climb/Accelerate to Max Speed and Altitude
6-7	200 nm Cruise Back
7-8	Descend to Sea Level
8-9	Reserves: Fuel for 30 Minutes at Sea Level at Speed for Maximum Endurance

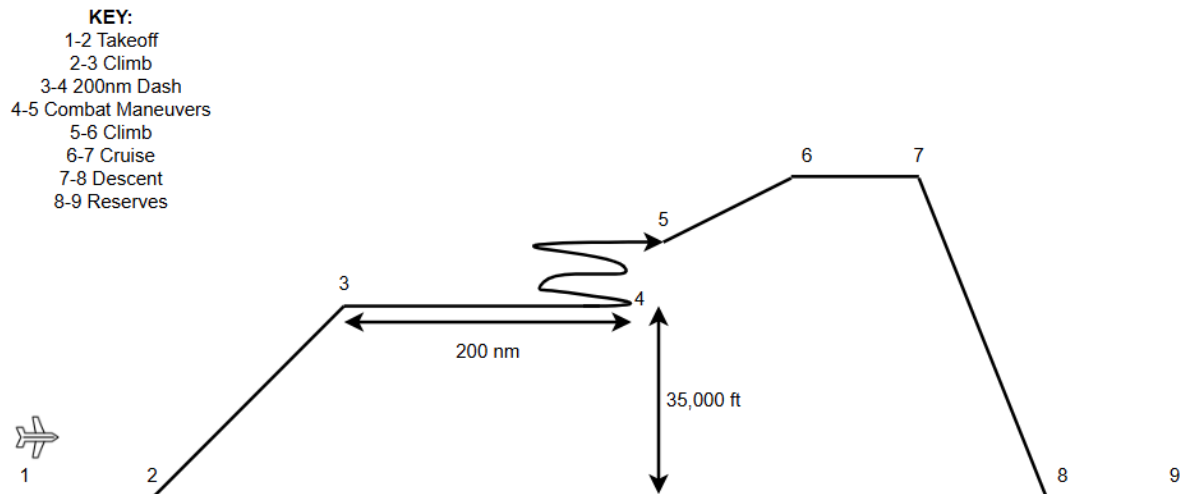


Figure 4.2.2: Point Defense Intercept Mission Profile

The mission begins with takeoff and acceleration to climb speed at maximum thrust, ensuring minimal response time. Upon reaching 35,000 feet, the aircraft must dash 200 nm at maximum speed, intercepting hostile targets before they reach critical assets or population centers.



Once within engagement range, the HDI must execute high-speed combat maneuvers, including sustained 360° turns at Mach 1.2 and Mach 0.9, before launching missiles. The aircraft must then climb and accelerate to optimal altitude and speed for an efficient return flight. The mission concludes with a 200 nm cruise back to base, followed by a descent to sea level and a 30-minute fuel reserve for safety. Given the high-energy demands of this mission, the HDI's propulsion system must deliver exceptional thrust-to-weight performance while maintaining fuel efficiency to extend operational reach.

4.2.3 Intercept/Escort Mission

The Intercept/Escort Mission evaluates the HDI's ability to escort and protect high-value assets, such as reconnaissance or transport aircraft, while maintaining full combat readiness. The mission profile follows a similar structure to the point defense mission but emphasizes extended escort operations rather than immediate interception.

The mission begins with a takeoff and rapid climb to 35,000 feet, followed by a high-speed dash to the escort area. Once on station, the HDI must maintain formation with the escorted aircraft for 300 nm at the lowest practical airspeed, ensuring continued protection while preserving fuel reserves. If necessary, the aircraft must be capable of engaging enemy threats while maintaining escort integrity. Following mission completion, the HDI will climb and accelerate to optimal speed and altitude, execute a cruise back to base, and conclude with a safe descent and a 30-minute fuel reserve at sea level.

Table 4.2.3: Intercept/Escort Mission Phases

Segment	Description
0-1	Take-off
1-2	Climb to 35,000 ft & accelerate to maximum speed
2-3	Cruise & dash out at maximum speed
3-4a	Climb/accelerate to optimum speed & altitude
3-4b	Escort for 300 nm at minimum practical speed (retain all weapons)
4-5	Descent to sea level
5-6	Reserves for 30 mins



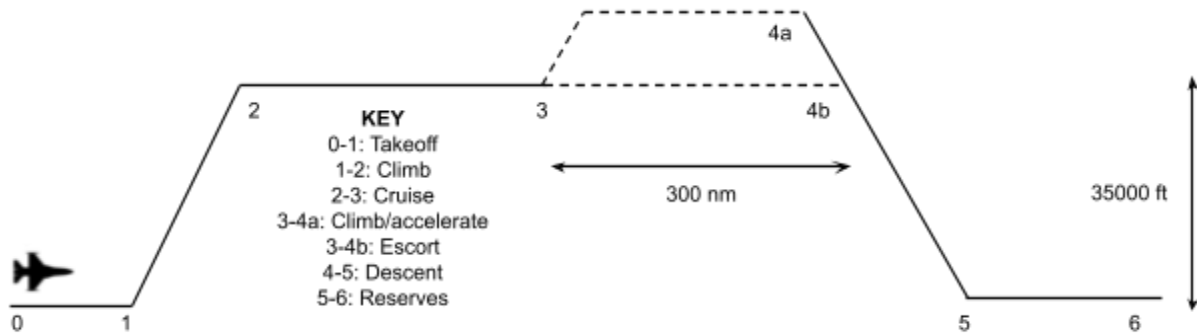


Figure 4.2.3: Intercept/Escort Mission Profile

5. Aircraft Design Approach

This section will summarize our team's approach for developing the final design we determined to be the optimal solution. We will begin by presenting the final design, then showing the complete process by which it was produced, along with each major milestone.

5.1 The TITAN

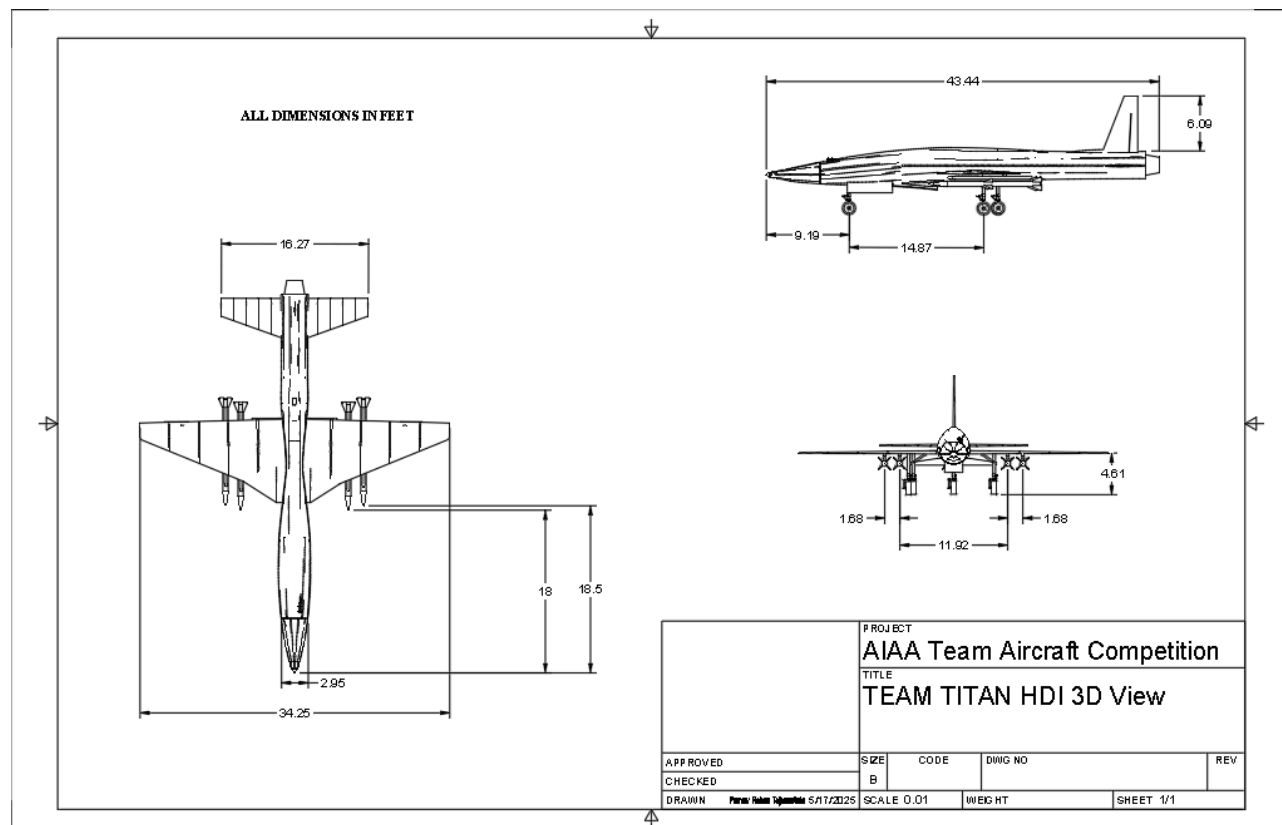


Figure 5.1.1: TITAN 3D View

The craft design that shares its name with our team, TITAN, is shown in Figure 5.1.1. TITAN's main concept was to be as low-weight as possible while utilizing well-understood aircraft concepts. It was also decided early on that the craft would be a UAV (unmanned aerial vehicle) to take advantage of weight savings, cost savings, and make an overall simpler design. The results of these fundamental concepts can be seen clearly in its fuselage, which has been optimized for low drag and minimal weight, and space. Due to the craft being unmanned, there was no space needed for a cockpit, and the space that would be otherwise reserved for it was replaced by the essential internal systems for the craft. With this design being required to hit speeds of Mach 1.6, a thin airfoil with a lower aspect ratio is used to ensure that the craft will be low drag, nimble enough to be maneuverable at high speeds, and take advantage of the resultant larger maximum thickness, which enables greater internal volume for systems. Lastly, TITAN is equipped with a single Honeywell F125-IN engine, ensuring simplicity, features 4 AIM-120 AMRAAM missiles, 2 on either side, and includes a 20mm M61A1 gun located at the nose-section of the aircraft.

5.2 Conceptual Design

The TITAN team began the conceptual design process by considering two fundamental classes of design: derivatives of existing aircraft and clean-sheet designs. After evaluating the complexity, time requirements, and risk associated with clean-sheet development, the team chose to focus on derivatives of proven aircraft to accelerate design maturity and reduce uncertainty. Several existing platforms were analyzed for suitability, including the T-38 Talon, F-5E Tiger II, and MiG-31. After assessing these candidates against the performance and cost requirements outlined in the Homeland Defense Interceptor RFP, the team selected the T-38/F-5 family as the most appropriate baseline due to their compact size, lightweight structure, and historical affordability. The final design evolved from a notional unmanned variant of the T-38, with major modifications including removal of the cockpit and life-support systems, integration of composite structural materials, and accommodations for modern avionics, GFE, and weapons systems. The switch to a single, relatively medium-bypass turbofan engine improved fuel efficiency while reducing weight and overall system complexity. These refinements allowed for a smaller, lighter configuration that meets all mission performance goals while remaining under the \$25 million flyaway cost cap mandated by the RFP.

5.3 Overview of Chosen Concept



To narrow down to our chosen design, we participated in a conceptual design selection with our advisors to compare proposed design concepts. The specific criteria that we focused on judging for each design were the simplicity of the design, the cost of materials, total weight, and whether the design hits the target parameters. During the meeting, we judged each craft based on these criteria, noting the benefits and tradeoffs of each design.

5.4 Initial Sizing

The first major category that we judged the feasibility of our two design classes was our weight. We began by calculating our desired takeoff gross weight, which is the weight that the aircraft starts each required mission at. This value was calculated by breaking down the total weight into multiple different weight components, starting with the crew, payload, fuel, and empty weight. Crew and payload were both given weights as defined by the design requirements, as our aircraft is a UAV, and our weapon payload is determined by the RFP. To calculate the unknowns, we first used geometry sizing. We chose our overall length to be 44 and our target weight to be 12000 lbs. After that, we estimated fuel fractions on different stages of the mission profile, such as start up, takeoff, climb, loiter, and landing. We then calculated the total fuel required, including reserve fuel. We then put all calculated weights into a mass properties calculator to find the overall weight of each concept.

6. Aerodynamics

6.1 Airfoil Selection

The wing design process was influenced heavily by existing aircraft such as the T-38 Talon and F-5E Tiger II, both of which feature wing shapes optimized for high-speed intercept and training missions. The T-38's use of a thin, low-drag airfoil and swept wing geometry provided a reference for achieving efficient supersonic performance in our design. Meanwhile, the F-5E Tiger II's structural simplicity and compact wingbox served as a model for manufacturability, internal volume efficiency, and control surface packaging. These proven designs demonstrated the balance of speed, maneuverability, and payload capability that we aimed to replicate in our interceptor configuration.

The selection of the airfoil was based on careful consideration of the Homeland Defense Interceptor's mission envelope, including loiter efficiency, climb rate, high-speed dash performance, and weapons integration. At a desired max of Mach 1.6, a supercritical airfoil was required to manage the transonic drag rise at high speed. Three primary airfoils were evaluated during early concept phases: the NACA 0006, the TsAGI S-12, and the NACA 63-412. The NACA 0006 was initially selected for consideration due to the symmetrical nature and improved



performance of the airfoil at supersonic flight. While it offered benefits in high-speed regimes due to its thin and symmetric profile, minimizing wave drag and supporting supersonic flight, it was ultimately not selected due to concerns about its low-lift behavior in subsonic phases of flight. As a symmetric airfoil, it requires a positive angle of attack to generate lift. While this can be mitigated through wing incidence angle or control surface deflection, preliminary configuration studies indicated that the required angles could result in unfavorable trim conditions and additional induced drag during low-speed operations such as takeoff and landing. These factors posed challenges to mission flexibility, particularly for loiter and climb phases, where smoother low-speed performance was prioritized. The TsAGI S-12 airfoil was also considered, with its notably high performance under supersonic conditions seen through its use in the MiG-21 fighter aircraft. Although it demonstrated high effectiveness at supersonic speeds, its overall range of performance did not match the specific operational requirements of our aircraft based on the RFP. For example, the usage of the TsAGI S-12 airfoil is more suited for a delta wing design with its maximum operational efficiency in supersonic speeds, as seen in the MiG-21, which is capable of reaching Mach 2.05 and operates mostly at supersonic speeds. Therefore, the use of that airfoil wouldn't be as effective as other options for all of the missions given in the RFP.

The NACA 65A004.8 airfoil, which was used on both the T-38 Talon and the F-5E Tiger II, was also reviewed during historical benchmarking. Its extensive wind tunnel validation, proven in-service reliability, and performance up to Mach 1.6 made it a viable candidate. However, it was ultimately not selected due to its specific tailoring for higher sweep angles and more slender wings than those used in our design. This can be seen in the aspect ratio of the F-5E Tiger being slightly higher than our aspect ratio using a different airfoil, which shows that our selection caters more specifically to the missions that TITAN will execute as given in the RFP, giving our aircraft increased maneuverability at high speeds. In addition, the NACA 64A204, the final airfoil chosen, offered a more straightforward implementation within our schedule while still achieving the necessary critical Mach number and thickness-to-chord ratio targets. Nevertheless, the 65A004.8 remains a strong alternative that may be re-evaluated if future design iterations allow.

Ultimately, the NACA 63-412 was selected, a cambered airfoil used on high-subsonic and low-supersonic platforms for its superior lift characteristics, good transonic drag behavior, and stable performance up to Mach 1.6. This airfoil allowed for smoother transitions during loiter and climb, as well as better efficiency during lower-speed



operations such as takeoff and landing. Unfortunately, initial design showed the shortcomings of the NACA 63-412, as the airfoil was too thick to operate at the speeds necessary for the mission.

Therefore, we chose to modify the airfoil selection again, and the NACA 64A204 airfoil was selected due to its favorable thickness-to-chord ratio of 4%, positioning it well for supersonic flight while maintaining enough lift at subsonic speeds. The thin profile of the airfoil reduces the wave drag experienced while the structure is still kept stable. Having a maximum thickness of 2.96 inches, it was a great compromise between efficiency and practicality. The thickness ratio is suited for flying in the supersonic regime, in addition to providing sufficient internal space for structural elements of the aircraft. The thickness-to-chord ratio also achieves a critical Mach number of approximately 1.6. This calculation includes the sweep angle, which was initially set at 45 degrees for improved supersonic efficiency, though current geometry reflects a 21.6° sweep. The calculated critical Mach number meets the required speed of Mach 1.6, set out by the RFP.

6.2 Wing Design

The wing configuration was developed to balance aerodynamic performance with the Homeland Defense Interceptor's compact form and mission-specific speed requirements. A wing loading of 90 lb/ft^2 was used as a design target, this value was based on aerodynamic reasoning and reference aircraft such as the F-5E Tiger II, which operates with a higher wing loading than the T-38A's (approximately 70 lb/ft^2). This slightly elevated wing loading supports a compact design with reduced span and structural weight, in line with the HDI's short-duration supersonic intercept mission. Using a gross weight of 12,000 lb, this yielded a required wing area of 133.33 ft^2 . From there, a wing aspect ratio of 3.5 was chosen, which falls within the range used by comparable supersonic aircraft and helps balance induced drag (at subsonic loiter) and wave drag (at Mach 1.6).

This geometry produced a wingspan of approximately 34 ft and a mean aerodynamic chord of approximately 8.96 ft. The initial wing sweep was set at 45° for favorable critical Mach number behavior, although our current CAD configuration uses a 21.6° sweep, a change that still maintains acceptable aerodynamic and structural performance. The change in critical Mach number due to this change in sweep angle is small enough to be considered negligible, and therefore, the sweep angle being set at 21.6° is beneficial for other reasons, such as



improving structural stability and subsonic performance. Overall, the wing planform was shaped to support high-speed flight while providing sufficient internal volume for fuel, structural members, and control systems.

6.3 Drag Analysis

Wave drag represents one of the most significant challenges in supersonic aircraft design, emerging as airflow exceeds Mach 1. In contrast to subsonic flight, in which pressure disturbances propagate in all directions, at supersonic speeds, these disturbances become trapped within a Mach cone at the back of the aircraft. As a result, shock waves form at multiple locations on the airframe, particularly at leading edges, wing-to-body junctions, and other discontinuities on the surface. As the local flow of air exceeds Mach 1, normal shocks form on the upper surface of the wing, causing a steep increase in pressure drag. As the aircraft speed increases above Mach 0.8 (transitioning into transonic), these shocks become more intense and move backward on the wing. Wave drag also occurs when the aircraft is traveling purely at supersonic speeds, as the shock waves develop along the leading edges, causing heavy drag due to a pressure gradient. Lastly, wave drag can occur simply due to the interaction of the shock waves on different aircraft components, which is why it occurs at junctions on the aircraft to further increase drag. All of these phenomena can be seen in the diagram below.

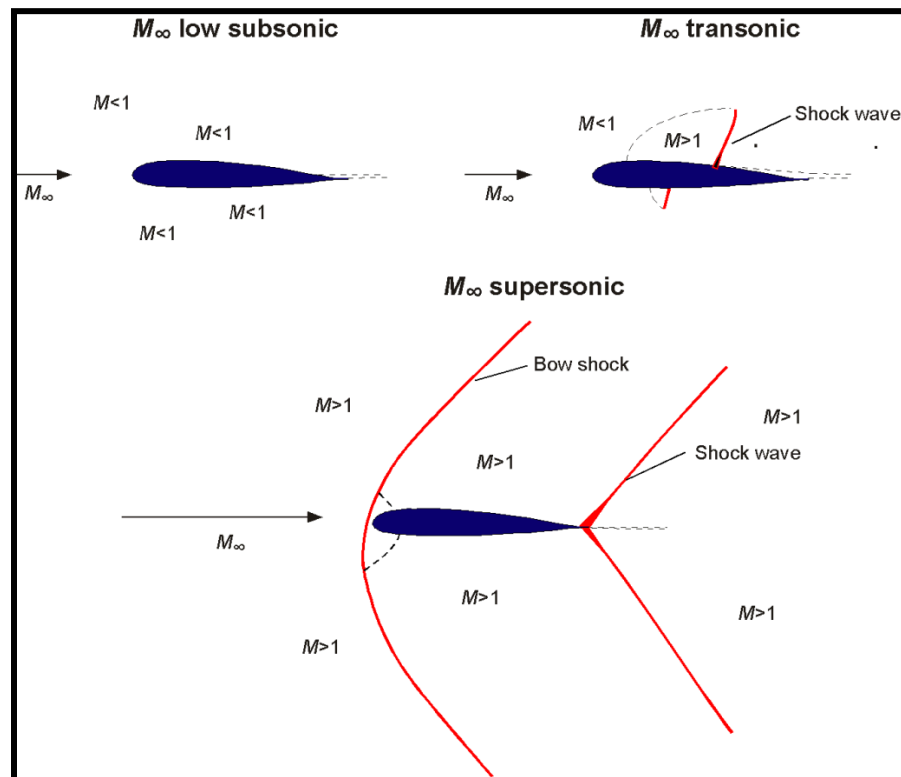


Figure 6.3.1: Development of wave drag in transonic and supersonic flight

Our aircraft includes several established methods for reducing wave drag at transonic and supersonic speeds. The 21.6-degree wing sweep angle delays the onset of wave drag by reducing the effective airfoil thickness with respect to the oncoming airflow. This method, originally developed in early supersonic aircraft design, reduces the effective chord ratio at the root by a factor of 0.929, a result of calculations concerning TITAN's dimensions. The cross-sectional area profile of the plane also helps reduce wave drag, with a gradual change in cross-sectional area along the direction from nose to tail. This makes use of the idea of the fuselage area rule, which describes how it is necessary to arrange the cross-sectional area of the plane by smoothly changing it. As described by the rule, TITAN's fuselage is narrowed at the wings, and the overall change of the cross section from the nose to the tail is made into a smooth transition, both of which help avoid the formation of strong shock waves and therefore avoid significant wave drag effects. This technique diminishes the shock wave intensity and its associated drag, a practice employed by many supersonic aircraft. The thin airfoil selection also provides a brace against the effects of wave drag. The magnitude of wave drag is proportional to the square of the aircraft's thickness ratio, and by that principle the 4% thickness ratio provided by the NACA 64A204 helps decrease the magnitude of wave drag. Finally, the smooth integration of the wing to the body helps reduce the onset of shock waves in those areas. Overall, the incorporation of these elements lowers wave drag significantly and helps the aircraft perform efficiently at supersonic flight. This allows us to achieve the effective operation at the desired Mach number of 1.6 without imposing severe thrust demands or inflicting high amounts of fuel consumption penalties.

Our aircraft design also accounts for transonic drag management, a difficult aspect because there are mixed subsonic and supersonic flows simultaneously affecting different parts of the aircraft. Our solution to this problem was yet again heavily based on a thin airfoil design, which weakens the shock experienced in transonic flight. Another aspect of our design was the mid-wing placement, which optimized the location of the wing about the fuselage to prevent harmful interactions between the two that could contribute to more drag. These characteristics enable the aircraft to transition through the challenging transonic regime with minimal performance losses, thereby achieving successful acceleration to supersonic flight.



While these drag reduction strategies come from well-established principles of supersonic and transonic aircraft design, the key innovation in our approach was the application of Raymer's Design System (RDS) to optimize these features through iterations. The framework that RDS provided gave us accurate computational numbers, which we were able to use to optimize the various aspects of our wing design. Through RDS iterations, we were able to refine every aspect of TITAN's aerodynamics, such as sweep angle, fuselage contouring, and other methods to decrease wave drag effects. These integrated features with specific data points allow the aircraft to not only achieve efficient supersonic cruise but also transition through the transonic regime with less performance penalties. The result of these improvements is an aircraft that has significant improvements reducing wave drag at supersonic speeds while maintaining the maneuverability and performance capabilities at low speeds, such that it can effectively complete all of the missions desired in the RFP.

7. Propulsion and Performance

7.1 Propulsive Requirements

Based on the RFP, the engine selected for the Homeland Defense Interceptor (HDI) must provide adequate installed thrust to support all phases of the design missions, including climb, cruise, dash, and maneuvering engagements. The propulsion system must be capable of supporting the aircraft's electrical and hydraulic requirements, providing 50 kW of power for onboard systems. Additionally, the environmental control and avionics liquid cooling systems must be allocated 2% of the engine's mass flow. To ensure cost-effectiveness and ease of integration, it is highly desirable that the selected engine is a non-developmental item, meaning it should be an engine that is currently in production or forecasted to be in production by 2027.

7.2 Engine Selection

7.2.1 Design Approach

The engine selection process for the HDI required careful evaluation of multiple factors, including thrust-to-weight ratio, fuel efficiency, operational endurance, and mission performance. One of the primary considerations was whether to utilize a turbojet or low-bypass turbofan. While turbojets offer high thrust and are optimized for high-speed flight, they generally suffer from higher specific fuel consumption and reduced efficiency



at subsonic speeds. Given the mission profile of the interceptor, requiring both high-speed engagement and sustained operational endurance, a low bypass turbofan emerged as the optimal choice.

Initially, with no other constraints or requirements established, the General Electric F414 was used as a placeholder due to its status as a proven, high-thrust, low-bypass turbofan. However, as the HDI design matured, it became clear that the F414-GE-400 exceeded the power requirements for a lightweight, single-engine interceptor. As shown in Figure 7.2.1, the F414-GE-400 delivers a thrust-to-weight ratio of 9.0, outperforming the F125-IN engine's ratio of approximately 6.8; this advantage is accompanied by significant fuel consumption trade-offs. According to Figure 7.2.2, the F414-GE-400 provides 22,000 lbf of thrust with a specific fuel consumption (TSFC) of 1.85 lb/lbf-hr, compared to the F125-IN's 9,500 lbf at 1.91 lb/lbf-hr (Deagel, 2025). Despite the F414's greater raw thrust, its higher fuel demands and greater mass make it less compatible with the HDI's mission requirements, which emphasize endurance, fuel efficiency, and rapid, affordable response. The F125-IN offers a more balanced propulsion solution for the operational and structural goals of the aircraft.

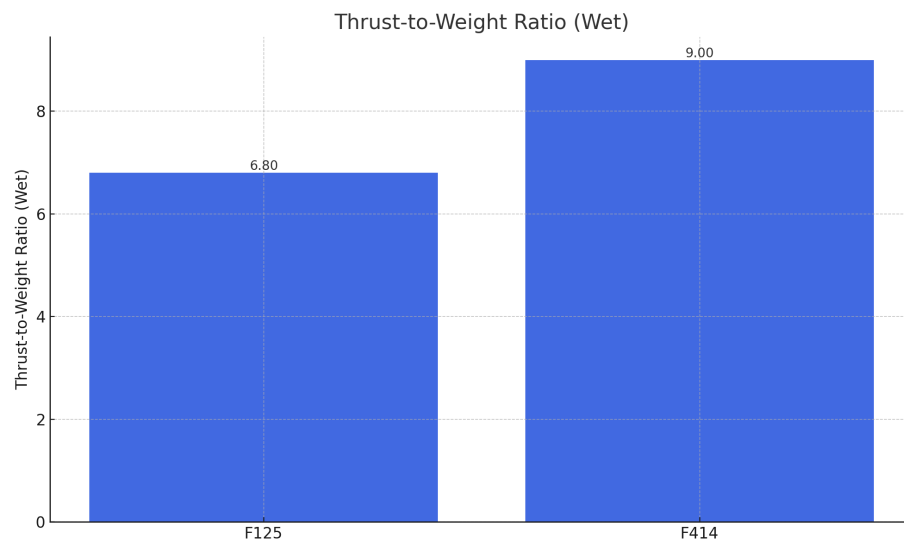


Figure 7.2.1: Wet thrust-to-weight ratio comparison between the F125 and F414 engines



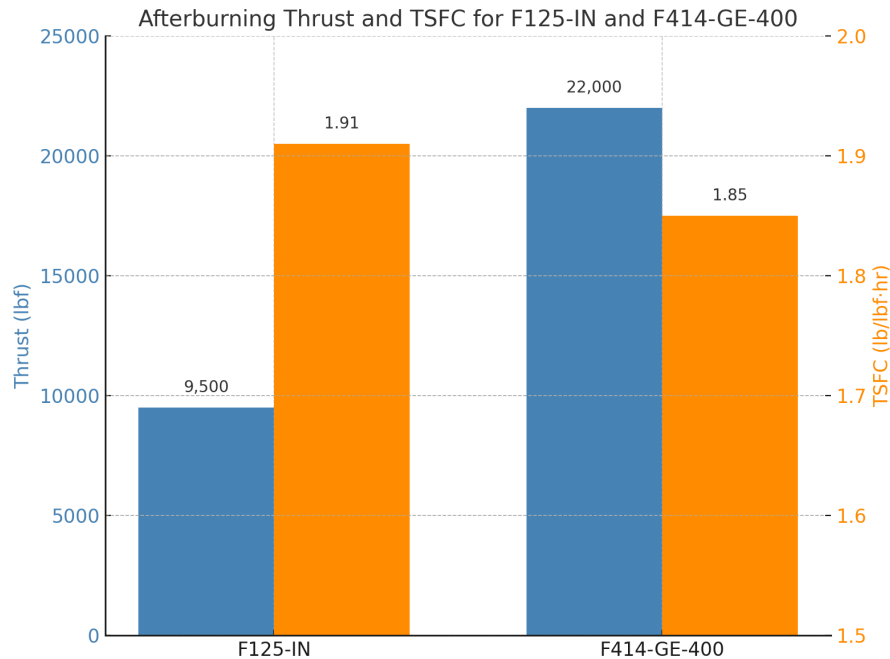


Figure 7.2.2: Afterburning thrust versus TSFC for F125 and F414 engines

These trade-offs ultimately led to the replacement of the F414-GE-400 with a more balanced propulsion system. The F125-IN was selected for its ability to deliver adequate thrust while maintaining better fuel efficiency in line with the HDI's mission needs. According to performance data, the F414-GE-400 produces 13,000 lbf of dry thrust with a TSFC of approximately 0.84 lb/lbf-hr, while the F125-IN provides 6,280 lbf of dry thrust at a TSFC of 0.775 lb/lbf-hr (Deagel, 2025). Under afterburner, the F414-GE-400 delivers 22,000 lbf at a TSFC of 1.85, and the F125-IN achieves 9,500 lbf at 1.91 lb/lbf-hr. These values are summarized in Table 7.2.1, which provides a side-by-side comparison using rated conditions rather than variable thrust points. Although the F414-GE-400 offers greater absolute thrust, the F125-IN offers a more efficient balance of fuel usage and power that aligns better with the endurance, affordability, and size requirements of a single-engine interceptor.



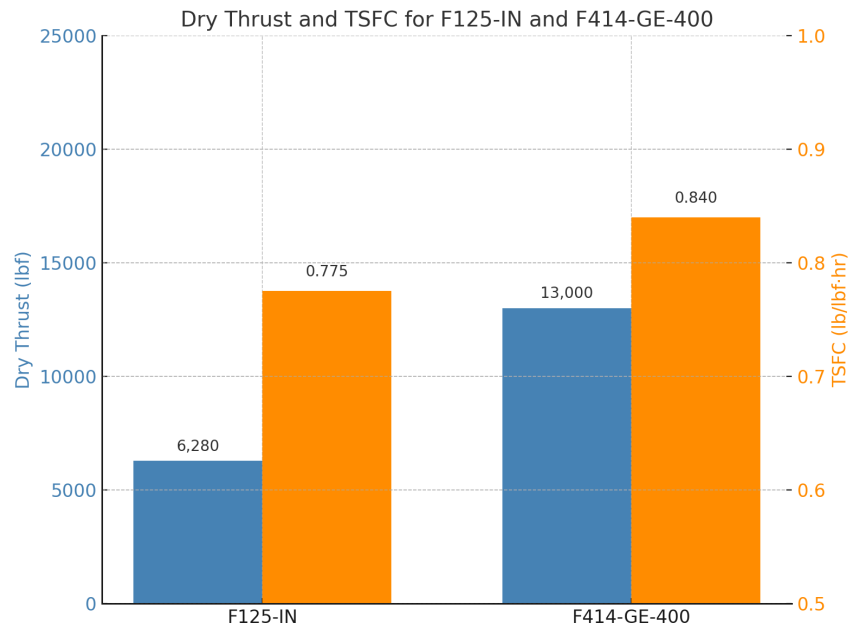


Figure 7.2.3: Dry thrust versus TSFC for F125 and F414 engines

Figure 7.2.3 illustrates the frontal area comparison between single- and twin-engine configurations for the F125-IN and F414-GE-400 engines. Although the F125-IN has a slightly larger frontal diameter than the F414-GE-400 when compared one-to-one, the total frontal area of the aircraft remains lower in the single-engine configuration. Internal engine integration allows the HDI to maintain a compact fuselage cross-section and minimize flow disturbances. In contrast, a twin-engine configuration would require a wider fuselage and larger structural accommodations for dual inlet and exhaust systems, increasing total wetted area and interference effects. By using a single F125-IN engine, the HDI achieves a more aerodynamically efficient layout that better supports its requirements for sustained supersonic speeds, reduced drag, and improved fuel economy over long missions.



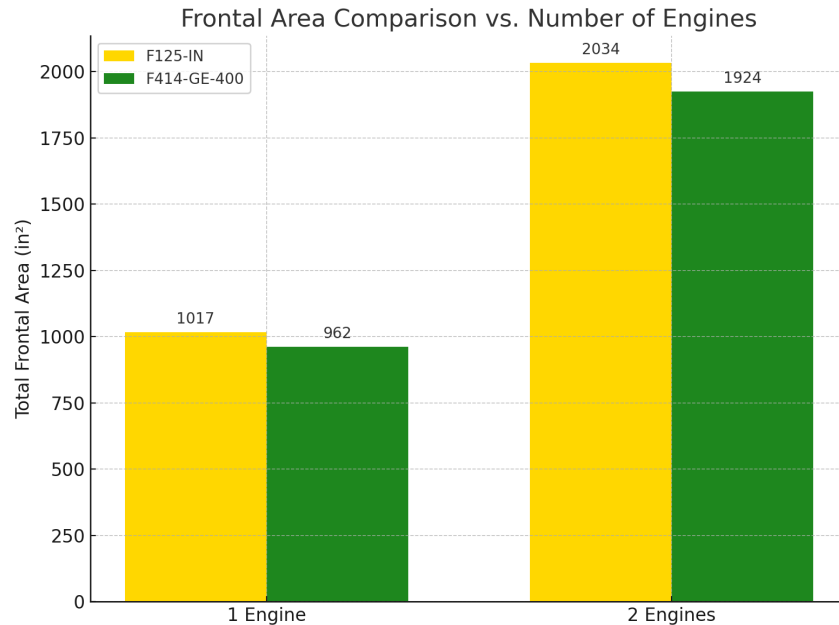


Figure 7.2.4: Frontal area comparison versus number of engines for the F125 and F414

Ultimately, the F125-IN engine was selected because it delivers the necessary power, endurance, and efficiency required for the Homeland Defense Interceptor’s missions. This choice ensures that the aircraft meets all operational requirements, including sustained supersonic speeds, maneuverability, and fuel efficiency, while maintaining a lightweight and effective propulsion system.

Table 7.2.1: F125 Key Characteristics

Characteristic	Value
Dry Weight	1360 lbm
Dry Thrust	6280 lbm
Max Thrust (AB)	9500 lbm
TSFC (dry)	0.775 lb/lbf/hr
TSFC (AB)	1.910 lb/lbf/hr
Bypass Ratio	0.472 : 1
Diameter	36 in
Length	124 in



7.3 Engine Placement

The placement of the F125-IN low-bypass turbofan within the Homeland Defense Interceptor was determined based on aerodynamic efficiency, structural integration, and mission requirements. Given the aircraft's lightweight and compact design, the engine was embedded within the fuselage to minimize drag and maintain a streamlined profile. However, the T-38-inspired airframe introduced constraints on engine placement due to its narrow fuselage and low-profile design, which limited available space for integration while requiring careful consideration of weight distribution and aerodynamic efficiency. These factors led to the decision to position the F125-IN at the back center of the aircraft body, ensuring an efficient exhaust flow path, proper balance, and reduced aerodynamic disturbances.

From the outset, it was determined that the F125-IN bypass turbofan engine should be positioned at the back center of the aircraft body to ensure a well-balanced center of gravity and an efficient exhaust flow path. This placement minimizes aerodynamic disturbances, simplifies exhaust routing, and improves overall stability. Additionally, by embedding the engine within the fuselage, the Homeland Defense Interceptor successfully balances aerodynamic performance, structural integrity, and propulsion efficiency while adhering to the constraints imposed by the T-38-inspired airframe.

7.4 Inlet and Nozzle Design

(pg. 285) As the chosen engine is a turbofan design, the incoming air must be slowed from above Mach 1 to no more than Mach .4-.5 [1]. Therefore, the inlet duct must slow down the incoming air enough to ensure that it reaches this speed to prevent it from stalling the engine. Before we get to geometry, one of the most important determinations for the effectiveness of the inlet is its location. There are many possible configurations, and each has its advantages and disadvantages. For this design, our team assessed 3 separate locations to determine which would be the best option for our aircraft. The locations we focused on were 'over-fuselage', 'chin', and 'side' (Figures 7.4.1-7.4.3).



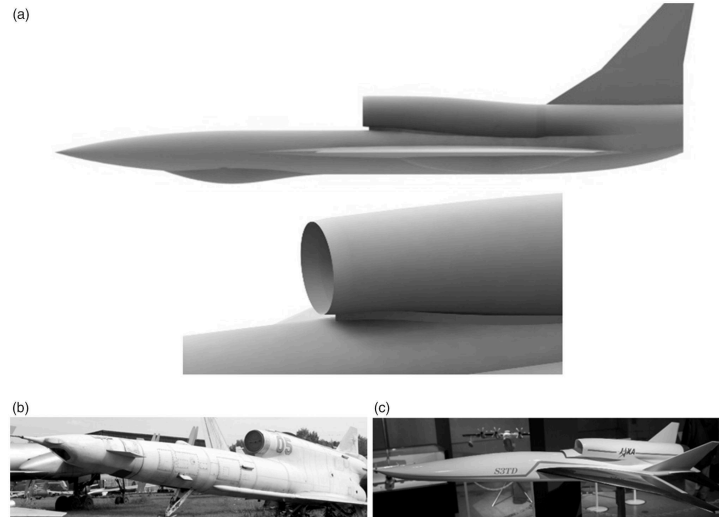


Figure 7.4.1: ‘Over-fuselage’ Inlet



Figure 7.4.2: ‘Chin’ Inlet





Figure 7.4.3: ‘Side’ Inlet

One of the rationales for choosing these 3 designs was the number of inlets. As we only have a single engine, two separate inlets are not completely necessary, and opting for a single inlet would help in reducing drag and simplifying geometry. For this reason, the ‘chin’ and ‘over-fuselage’ locations were assessed. Famously, the F-16 utilized a ‘chin’ design. Besides making geometry simpler, it also provides a completely unaffected flow for the inlet and the rest of the craft, such as the wing and a boundary layer formed by the fuselage surface. It is also great at high angles of attack due to its being under the fuselage. However, it creates a multitude of issues with the placement of the front landing gear. Placing the gear in front of the inlet would obstruct flow, and placing it behind requires a longer cowl to carry it. Therefore, we decided against this design to keep the overall craft as simplistic as possible. For the over-fuselage design, it had similar advantages to the chin inlet, but unfortunately, it suffers at high angles of attack due to the forebody blocking flow, which could lead to stalling. Therefore, the ‘side’ design was determined to be the best option. Side inlets can still provide the advantage of short ducts and relatively clean air if placed adequately. As we are using only a single engine, we designed a split duct design, ensuring that when both halves of the inlet flow meet, they do not stall the engine by keeping them separate up to the engine face.

7.5 Engine Integration

7.5.1 Engine Mounting System

The HDI’s engine is mounted along the rear fuselage centerline, utilizing a dedicated thrust frame structurally integrated into the rear fuselage and tied directly into the aircraft’s primary structural ribs. This



configuration provides a direct load path for transmitting axial and bending loads into the airframe, enhancing structural efficiency and preserving aerodynamic balance. The mounting system is designed to withstand +7 and -3g load factors and with a factor of safety of 1.5, as specified in the structural performance requirements of the RFP. All components are sized with a minimum factor of safety of 1.5 applied to ultimate loads, ensuring durability and structural reliability during peak loading events.

To protect sensitive avionics and airframe components from engine-induced vibrations, vibration isolation bushings are integrated at critical mount points. These features reduce the transmission of cyclic mechanical loads, minimizing fatigue-related degradation and contributing to the long-term service life of surrounding systems.

To satisfy the RFP's maintainability requirements, the mounting system includes removable access panels positioned along the aft fuselage. These panels allow maintenance personnel to inspect and service the engine mounts, vibration isolators, and connection interfaces without requiring full disassembly of the fuselage structure. During removal, the engine is extracted vertically using an overhead lift system after detaching mechanical and fluid connections, including quick-disconnect hydraulic fittings, dry-break fuel couplings, and color-coded electrical harnesses. Precision alignment pins guide reinstallation to ensure repeatable structural integration. This design enables streamlined engine replacement procedures, supports efficient turnaround times, and ensures maintainability even in austere or forward-deployed operating environments.

A combination of titanium and aluminum alloys is employed in the mounting assembly, with material selection based on localized structural and thermal requirements. Titanium is used in high-stress and high-temperature zones for its exceptional strength-to-weight ratio and heat resistance. Aluminum alloys are applied in peripheral, lower-load structural elements to reduce weight. Steel is reserved for high-strength fasteners and concentrated load transfer points. This material distribution provides a balanced solution that meets structural performance goals while minimizing excess mass.

7.6 Takeoff and Landing

The selected F125-IN engine provides over 9,000 lbf of thrust with afterburner, enabling rapid acceleration and a short takeoff roll consistent with HDI's requirement for conventional runway operations under 8,000 feet. The aircraft's low weight and efficient thrust-to-weight ratio contribute to improved takeoff performance. The thin airfoil



and moderate wing area also reduce drag, allowing a stable takeoff profile. Landing is performed under idle or near-idle thrust and supported by aerodynamic braking and conventional landing gear systems. The airframe is designed for runway-based operations with no requirement for carrier or vertical takeoff capabilities.

7.7 Climb and Cruise

The Homeland Defense Interceptor (HDI) is designed to meet the mission performance requirements outlined in the RFP, which include a cruise ceiling of 35,000 ft and a maximum cruise speed of Mach 1.6. These constraints demand a propulsion and aerodynamic configuration that supports both high-speed dash and efficient sustained flight. The selected F125-IN engine, combined with a lightweight airframe and a low-drag NACA 64A204 airfoil, enables an efficient and structured ascent profile with a balance of thrust, fuel economy, and aerodynamic performance.

7.7.1 Climb

The HDI uses a single F125-IN engine, which provides 9,500 lbm of thrust with afterburner and 6,280 lbm in dry thrust. This engine was selected to deliver adequate acceleration and thrust-to-weight for conventional takeoff and intercept missions, while maintaining favorable fuel consumption for loiter and return phases. Given the HDI's relatively lightweight and compact design, the available thrust is expected to be sufficient to support a direct climb to cruise altitude without the need for step climbs or throttle modulation.

7.7.2 Cruise

Cruise is performed at a target altitude of 35,000 feet and a maximum cruise speed of Mach 1.6, consistent with the RFP. The aerodynamic configuration of the HDI was developed to support efficient high-subsonic and low-supersonic flight. The selected NACA 64A204 airfoil, with a 4% thickness-to-chord ratio, supports high critical Mach number performance while maintaining sufficient lift and internal volume. The wing sweep was reduced from an initial 45° to 21.6°, based on structural and integration trade-offs, while still preserving the airfoil's effectiveness at the design Mach number.

At cruise, the HDI operates without an afterburner to minimize fuel consumption and maximize loiter endurance. The F125-IN engine's dry TSFC (0.775 lb/lbf·hr) allows for efficient sustained flight, and its compact design helps maintain a favorable drag profile. Cruise is assumed to occur at constant altitude and constant Mach to



simplify operation, ensure throttle predictability, and comply with airspace regulations. The airframe is optimized for stability and controllability throughout cruise, with no requirement for in-flight refueling or external fuel tanks.

8. Weapons and Defensive Capabilities

8.1 Weapons Stations

The trade-off of the aircraft's mid-range missile carriage capabilities was carefully studied to abide by the project philosophy of swarm engagement using cheap and mass-produced light defense interceptors. Keeping the advantage of the affordable and high production volume of the aircraft fleet in mind, the numerical superiority in combat situations was assumed throughout. As a result, a trade decision was made to keep the minimum number of AIM-120 AMRAAM missiles while maintaining a high probability of kill (Pk).

Due to the lack of publicly disclosed information, we assumed that the AIM-120C/D is capable of a Pk of about 0.75-0.85. Ultimately, 0.75 individual Pk was assumed in the following calculations for a more conservative prediction. Then, to consider the “complement of all missiles missing”, the complement rule was used, employing the following equation:

$$Pk = 1 - (1 - p)^n$$

Where p is the kill chance of individual missiles, and n is the number of missiles launched at the target. Considering the case of two sequential launches of the later variants of the AMRAAMs on a single target,

$$Pk = 1 - (1 - 0.75)^2 = 93.75\%$$

a very high Pk of 93.75% was yielded. To complement situations where two sequential AMRAAM launches on a target are denied for various reasons, such as enemy deployment of chaff or use of ECM, and to ensure the possibilities of engaging on a maximum of two aircraft per mission, a trade decision was made for the equipment of four AMRAAMs per aircraft.



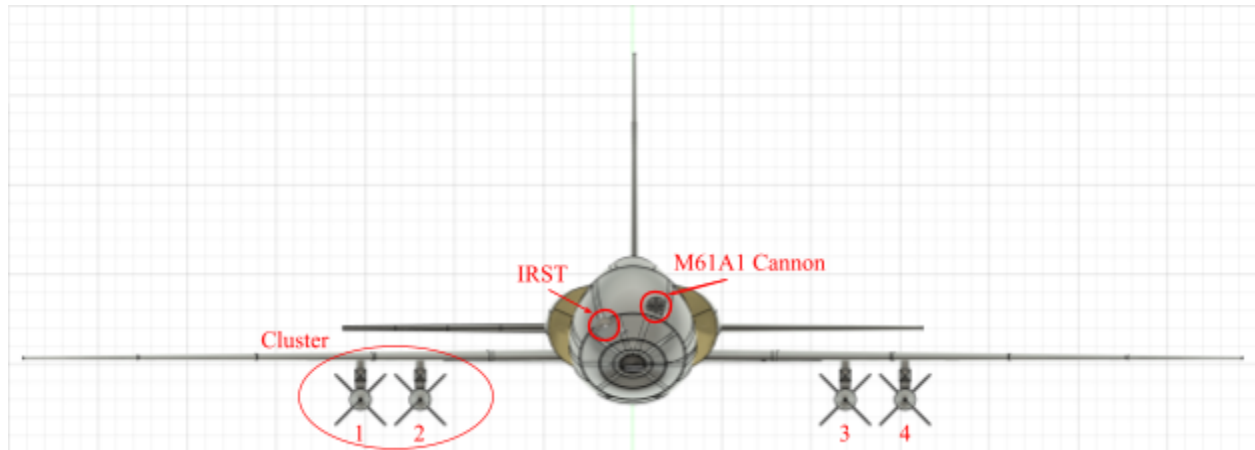


Figure 8.1.1: Weapons system and IRST positioning

In the configuration of the four AMRAAMs, standard LAU-128 missile launchers were selected for mounting on hardpoints beneath the wings, as shown in Figure 8.1.1. This approach was chosen over an internal bay carriage due to the significantly higher complexity, production, and maintenance costs associated with internal storage. Additionally, this decision aligns with the “swarm defense” philosophy, prioritizing rapid deployment and modularity.

Initially, a configuration using two double-launcher-equipped pylons—one under each wing—was considered to minimize the roll moment of the aircraft. However, the final design opted for a four-hardpoint configuration with two close-distance “launcher clusters” on each wing. This arrangement maintains a similar roll moment balance to the initial double-launcher setup but offers further advantages by reducing overall weight and drag, enhancing the aircraft’s aerodynamic performance.

The details of the mass properties of each part of the weapons system is included in the mass properties table, figure 10.4.2.



8.2 Integrated Gun

The aircraft is equipped with an M61A1 20mm cannon, as specified in the RFP. To minimize interference with theIRST (Infrared Search and Track) unit—positioned near the aircraft's nose—the cannon's location was carefully selected. Specifically, it was placed behind theIRST unit along the fuselage station direction. Both the cannon and theIRST were positioned as far apart as possible along the buttock line to reduce the impact of muzzle gas on theIRST's performance, without significantly compromising its field of view. TheIRST is located on the starboard side while the cannon is located on the port side, as shown in Figure 8.1.1.

9. Subsystems

9.1 Electrical Systems

The electrical systems of the Homeland Defense Interceptor (HDI) follow a simple and efficient layout scheme. The main part of the electrical system is the Auxiliary Power Unit (APU). This component is housed in the middle-front of the fuselage, in the underbelly region. This location allows for centralized wiring and easy diagnosis of other issues that stem from the center outward. Additionally, servicing for the APU will take minimal effort with easy access to the unit. A heavy reliance on maintenance is emphasized in order to minimize extra components and redundancy at times, to satisfy weight and cost requirements. Figure 9.1.1 shows the layout for the electrical system, with the internal structures omitted from the design file for easier viewing.



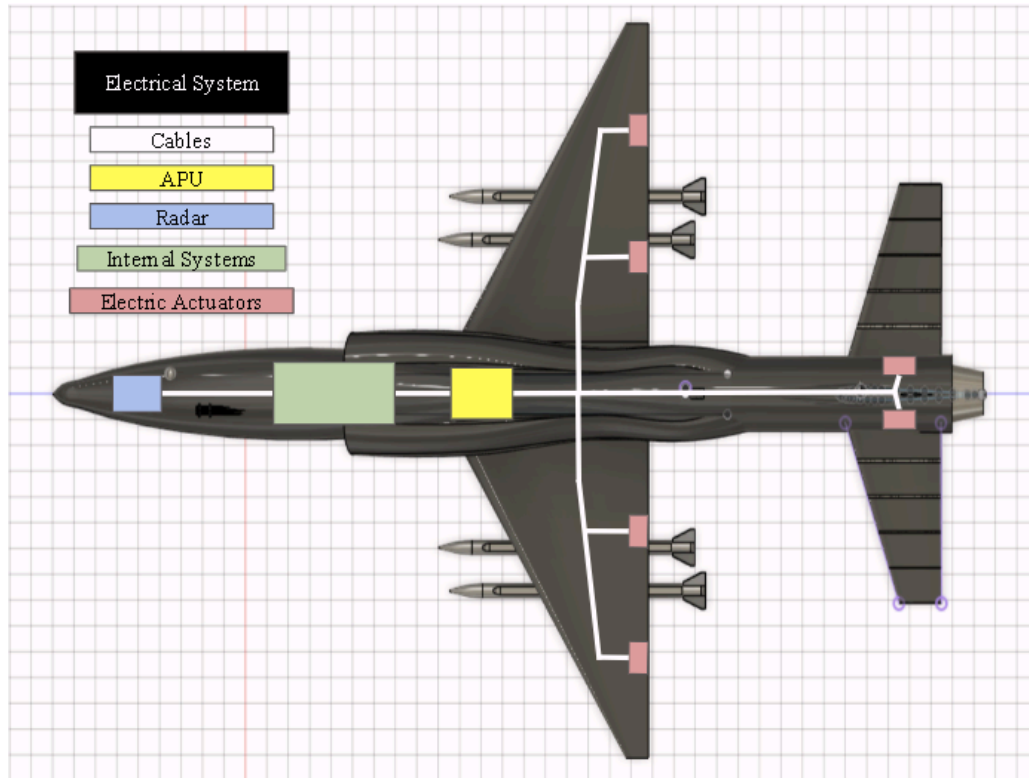


Figure 9.1.1: Electrical System Diagram

Other internal components have access to power through wires that run along plates and mounted surfaces inside the fuselage. The front of the fuselage has wires available for all of the internal components, including the front landing gear. The remaining power is distributed to the control surfaces, which are controlled by electric actuators. Both wings and the tail have wires running to them, in a branch or stem-like fashion. Each actuator has power available from smaller branches of wires that are connected to the main line, which runs along the wings and main part of the fuselage. The wing landing gears also have access to power through the main lines that run along the wings. As for the all-moving tail wings, the actuators are placed at the stabilator's root, which runs into the fuselage.

Another system not explicitly shown here is the digital fly-by-wire system. This system is implemented on TITAN to allow for easy remote piloting and full control from the on flight avionics to the control surfaces. While the system is not pictured, it runs very similarly to the electrical system, since it is based on electrical signals and wiring that will be set up together with other wires on TITAN.



Serviceability is also prioritized for TITAN, where maintenance ports for the electrical systems are spread out along the aircraft's body. Ports near the internal components where wiring will be allowed for easy fixes to any issues that may arise. This aspect is covered more later on in this proposal, where the maintenance port diagrams line up with the design of both the electrical system and the internal layout.

9.2 Fuel and Cooling Systems

The fuel and cooling systems of the HDI are carefully engineered to meet the demanding operational requirements of all three mission profiles: Defensive Counter-Air patrol, Point Defense Interception, and Intercept/Escort. These systems ensure sufficient internal fuel capacity for long-endurance loiter and high-speed engagement, while also maintaining safe thermal conditions for the propulsion system and mission-critical avionics. With self-sealing fuel tanks, JP-8 compatibility, and integrated environmental control subsystems, the HDI achieves a balance of performance, survivability, and maintainability suitable for NATO-aligned operations.

9.2.1 Fuel System Design

The HDI's internal fuel system is optimized for compact storage, survivability, and performance. The aircraft utilizes self-sealing internal fuel tanks to enhance survivability in combat environments, particularly in scenarios involving small-arms fire or fragmentation. These tanks are typically constructed using layered materials that include elastomeric compounds or other sealant technologies designed to react upon contact with fuel. This allows the system to automatically seal punctures, minimizing fuel leakage and reducing the risk of fire or critical system failure. The use of self-sealing tanks ensures the HDI can maintain fuel integrity even after sustaining damage during high-risk missions.

The selected fuel for the HDI is JP-8, the standard kerosene-based aviation fuel used by the U.S. military and NATO allies. The fuel selection process was directly informed by the operational demands of the HDI's three mission profiles. For the Defensive Counter-Air (DCA) patrol mission, which includes a 300 nautical mile outbound cruise, a four-hour loiter at 35,000 feet, and a 400 nautical mile return leg, JP-8's high volumetric energy density is critical for meeting endurance and range requirements without increasing internal fuel volume or airframe size.

In the Point Defense Interception and Intercept/Escort missions, while endurance is less of a constraint, JP-8 offers advantages in thrust availability, fuel efficiency during high-speed engagement phases, and logistical



interoperability. Its low freezing point (approximately -52°C) also ensures performance reliability in high-altitude and cold-weather environments, conditions likely encountered during interception and escort operations. Furthermore, JP-8's widespread use across U.S. and NATO facilities ensures compatibility with existing infrastructure, enabling rapid refueling and global deployability without the need for specialized handling or storage systems.

9.2.2 Fuel Port

The Homeland Defense Interceptor incorporates a Single-Point Ground Refueling (SPGR) system located on the left lower fuselage, slightly forward of the wing root. This placement is strategically selected to facilitate ease of access by ground support personnel, expedite aircraft turnaround times, and ensure safety by positioning the fuel inlet away from critical heat-generating areas such as engine exhausts and weaponry. While the Request for Proposal (RFP) does not explicitly dictate fuel port placement, this location aligns with operational expectations for NATO-compatible aircraft, optimizing ground compatibility and maintainability.

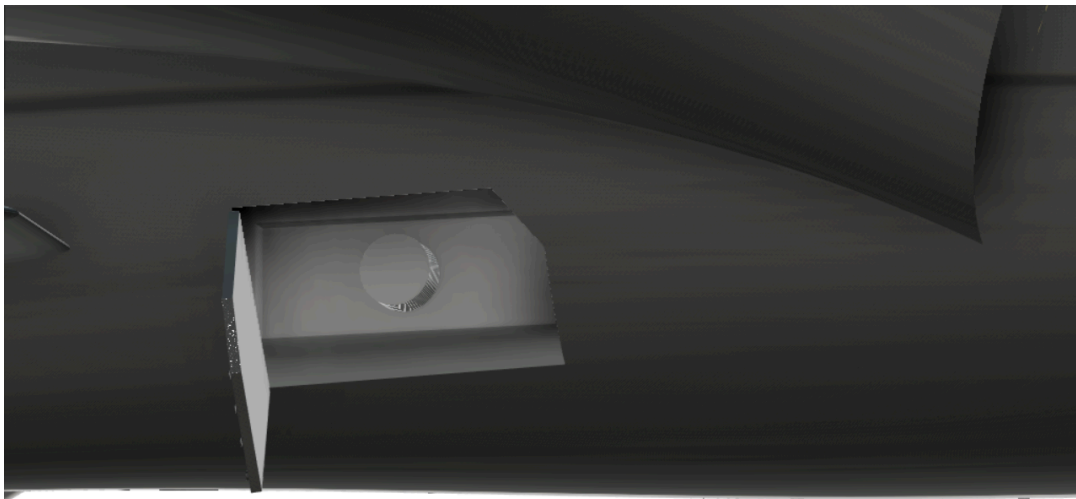


Figure 9.2.1: Fuel port

9.2.3 Cooling System

The HDI's cooling system is designed to ensure reliable performance across all mission profiles, including high-speed interception, extended patrol loitering, and remote avionics operation. As a remotely piloted aircraft, the



HDI must maintain thermal stability for mission-critical systems without onboard crew, necessitating a robust and efficient cooling architecture.

The primary system includes an oil cooling loop located near the engine bay, with airflow ducted through the fuselage to manage temperatures associated with engine operation. This setup supports consistent thermal regulation for lubrication and mechanical systems during high-thrust and extended cruise conditions.

Additionally, approximately 2% of engine compressor bleed air is allocated for use in the environmental control system (ECS) and for cooling avionics equipment. This approach is commonly used in military aircraft and enables internal systems to maintain operational temperatures under varying thermal loads without requiring large external radiators or heat exchangers.

To further manage localized thermal loads, especially in avionics and electronic systems, the HDI integrates fuel-based heat exchangers, positioned near the wing root. This location provides access to freestream airflow, enhancing heat rejection. The exchangers allow the aircraft to use fuel as a thermal sink, transferring excess heat into the fuel before combustion, minimizing added cooling system mass. This method supports compact system integration while reducing the need for dedicated thermal management components.

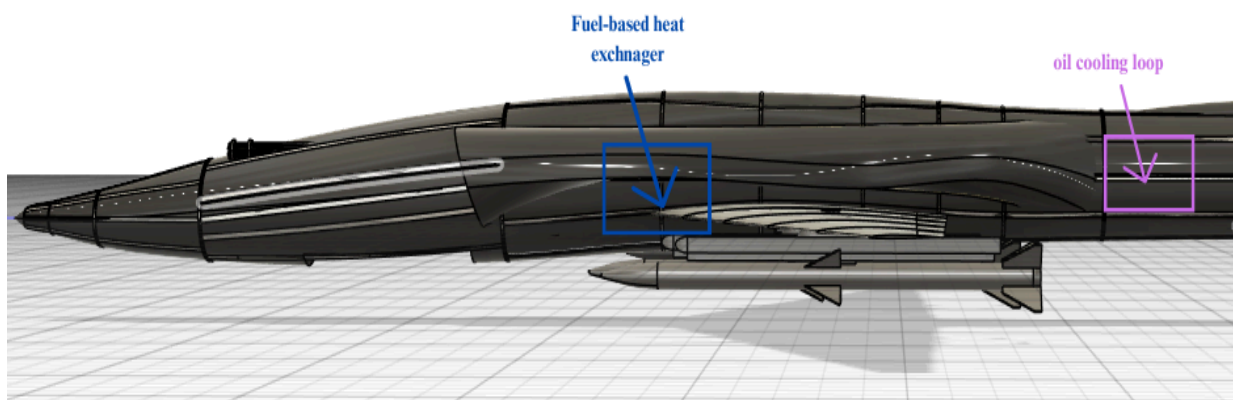


Figure 9.2.2: Proposed layout of engine and avionics cooling architecture for the HDI

Together, these subsystems ensure the HDI can support long-duration loiter, high-speed dash segments, and remote mission system operation without overheating. The cooling system is designed with efficiency, simplicity,



and system compatibility in mind to support rapid maintenance and sustained operation from both main operating bases and austere airfields.

9.3 Landing gear and recovery systems

The landing gear of the HDI has a tricycle configuration similar to many traditional interceptor designs, with a single landing gear in the nose cone region, and 2 landing gears placed under the wing. Each landing gear (fuselage and wing) contains a single tire, a wheel, a bracket, and a strut. Additionally, the wing landing gear possesses a drag strut for resistance to drag loads during takeoff and landing.

As for material selection, the tire will be composed of rubber while the rest of the gear will be made of Ti-6Al-4V, a standard alloy used in the industry. The placement of the gears in the figures below was selected so that the wing gears account for 85% of the load upon landing, with the nose gear accounting for 15%.

Structural analysis (FEA) was performed on this gear to determine if it could withstand landing on a land runway. While landing isn't required in the RFP missions, having landing capabilities makes the aircraft reusable. Reusability reduces long-term operational costs and extends service life by a significant amount. The gear should be able to withstand 2g's, with a factor of safety of 1.5, making it a total of 3g's (34,140 lbs)



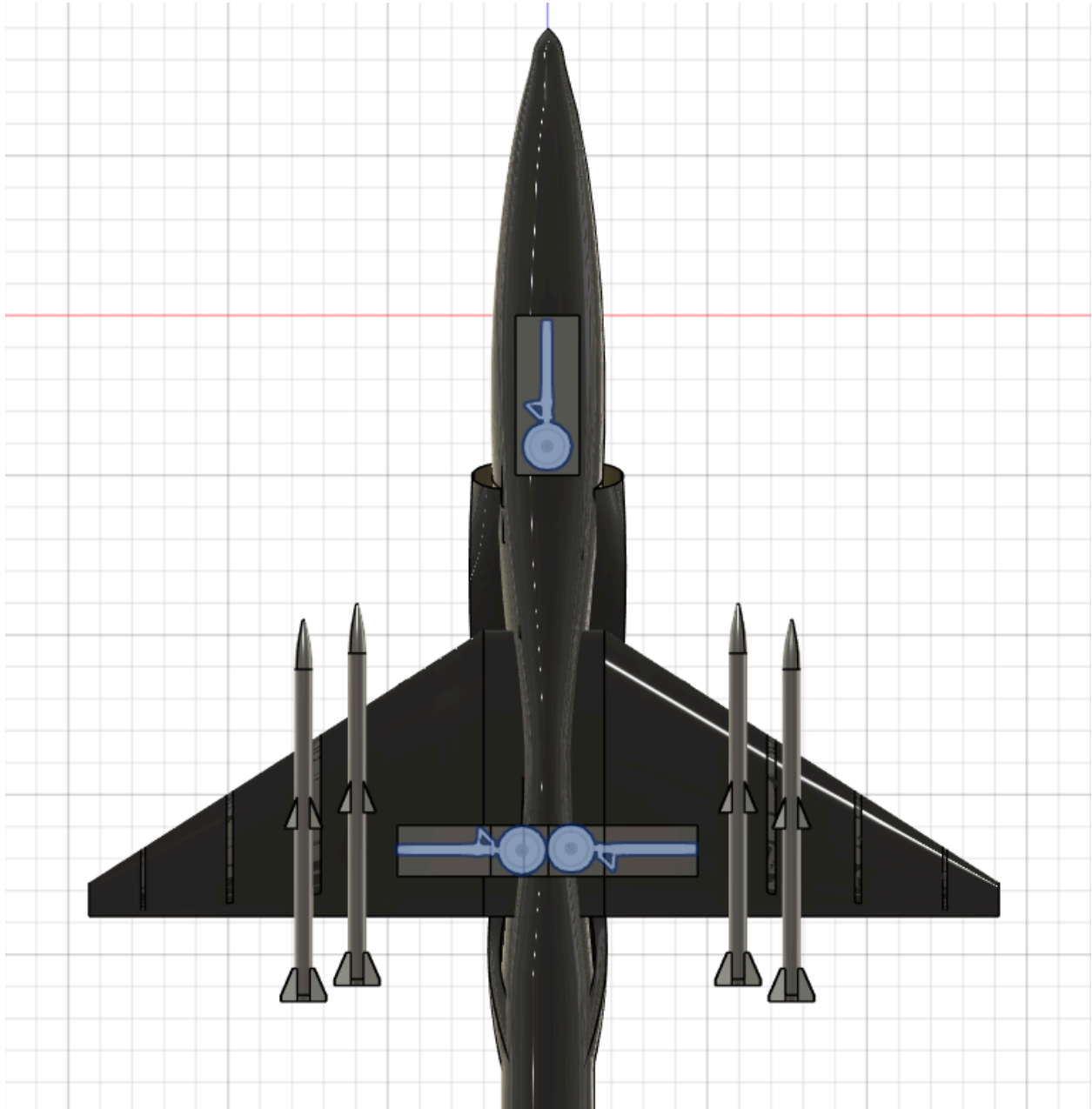


Figure 9.3.1 Landing gear placement (retracted view)



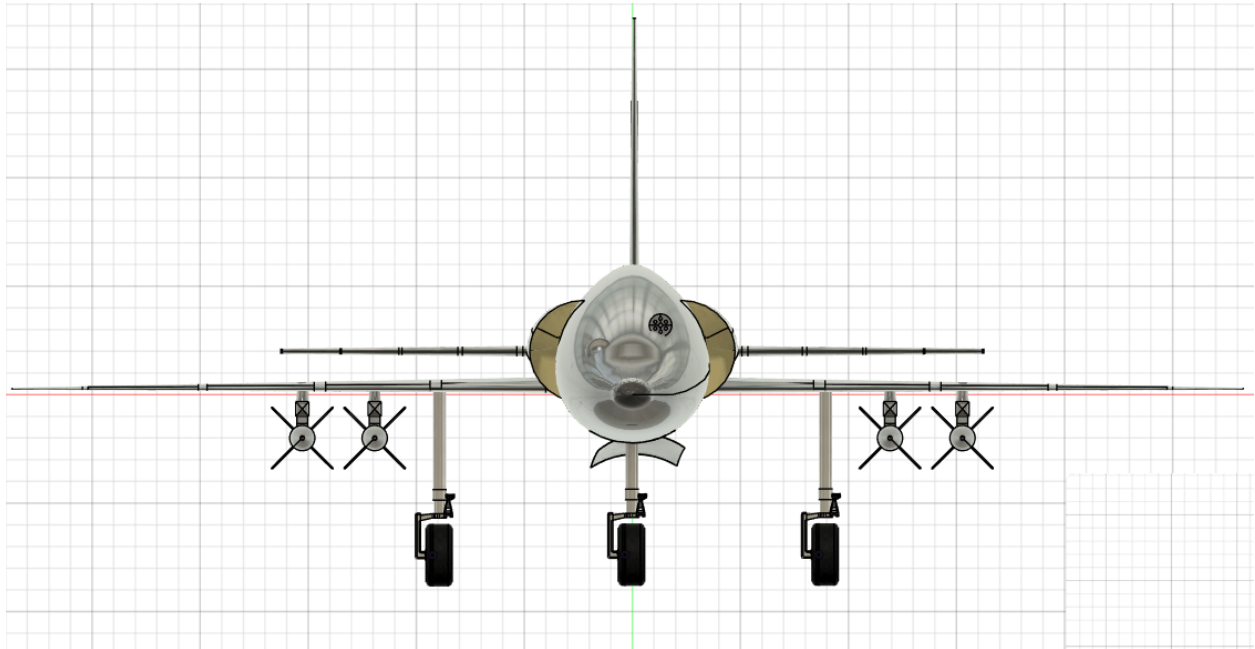


Figure 9.3.2: Landing Gear View

Table 9.3.1: Landing Gear Analysis Results

Nose Gear Coordinates	(9 ft, 0 ft)
Wing Landing Gear Coordinates	(25.76 ft, -4.5 ft), (25.76 ft, 4.5 ft)
Titanium 6Al-4V Yield Strength	880 MPa
Total force exerted on the gear	34,140 lbs
Force exerted on the wing gear	14510 lbs
Max Stress on wing gear	596.45 MPa
Max displacement of wing gear	0.758 inches
Force exerted on the nose gear	5,121 lbs
Max Stress on nose gear	210.32 MPa
Max displacement of nose gear	0.27 inches



Lastly, the nose gear retracts in the direction of the fuselage while the wing gears retract at the meeting point between the wing and fuselage, with some space cut out in the fuselage to fit the wing landing gears. There are rotating hinge gates at the landing gear bays to secure the bay during flight.

9.4 Avionics and Internal Systems

The Avionics and Internal Systems are configured so that the space from an empty cockpit can be maximized as much as possible. In our representative CAD model, most of these models simply represent the volumes based on the RFP requirements, which ensures that we design the aircraft to accommodate these spaces. The radar components are located in the nose section of the aircraft, which follows standard aircraft design and allows for access through a hinged nose cone (discussed later). Behind the radar is the electrical system provided in the RFP, which is dependent on the number of engines for the HDI. For TITAN, only one engine is incorporated, so there is only one of these system geometries.

Immediately behind the sole electrical system is the M61A1 20 mm cannon, with a small part of the barrel having access to the open air. The rest of the gun is located within the fuselage, towards the upper-middle area when observing a side profile (seen below in figure 9.4.1). The landing gear (not pictured) is below the gun. The rest of the components are separated by a floor, with an upper deck and lower deck to optimize accessibility and organization of the layout. Below are side views of the front half of TITAN's airframe, without the fuselage, to show the internal system layout.

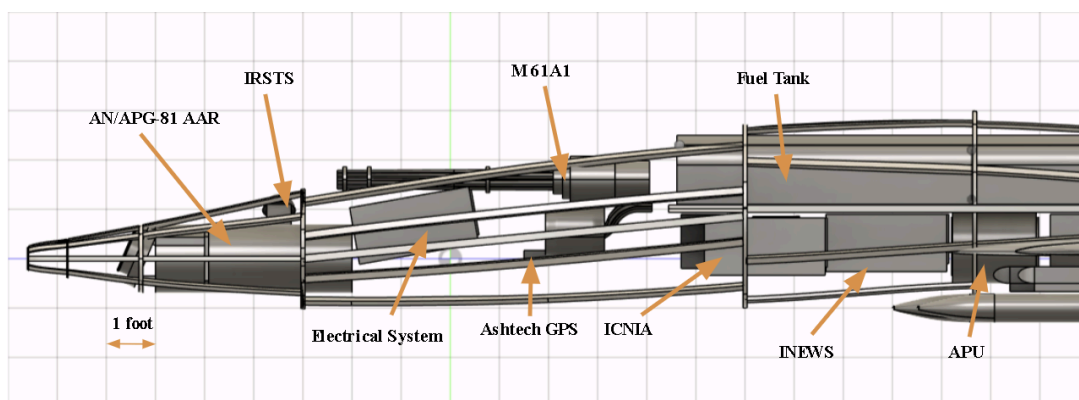


Figure 9.4.1: TITAN Right Side View



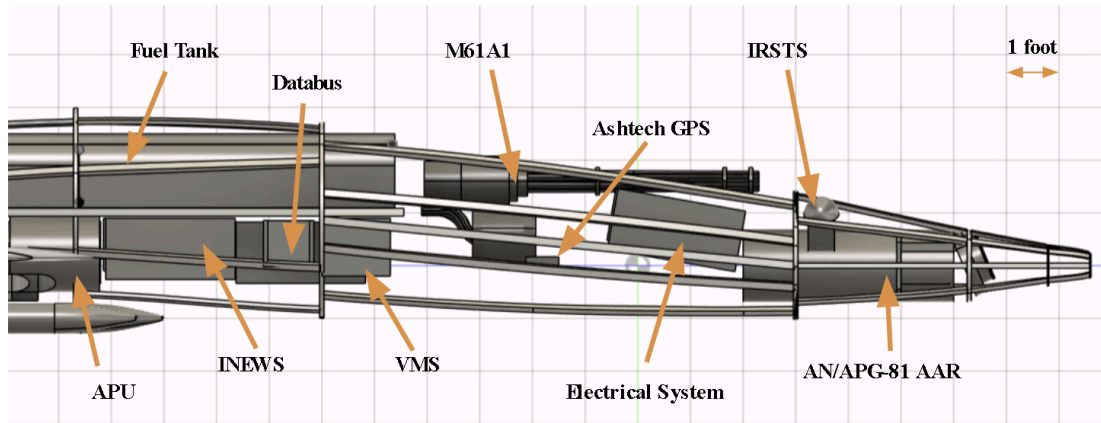


Figure 9.4.2: TITAN Left Side View

This layout works around the compact nature of our fuselage, and components are placed based on space constraints and servicing concerns. Later on, maintenance is further discussed and more detail is provided on how this layout allows for easy servicing and single-layer direct access for all internal components.

10. Structures and Mass Properties

10.1 Material Selection

For a craft design such as this, the chosen material must successfully balance total weight, cost, and workability. In addition to these overarching considerations, we also considered the loads that a supersonic aircraft would encounter, and ensured that both requirements of G-loads of +7 and -3 with a 1.5 factor of safety are being reached. Typically, most military craft, especially fighter jets, use some sort of metallic alloy, commonly aluminum or steel alloys. We narrowed our choices down to 6 different materials: 3 aluminum alloys, 2 titanium alloys, and a carbon fiber composite.

Table 10.1 Table of Materials

Material	Ultimate Tensile Strength (MPa)	Tensile Yield Strength (MPa)	Shear Strength: (MPa)	Young's Modulus (GPa)	Mass Density (lb/in ³)	Melting Point (°F)	Cost (\$, 2024, per lb)
Aluminum 7068	710	683	365	73.1	0.103	888.8	\$1.15-\$2.04
Aluminum 7075-T6	572	503	331	71.7	0.102	890.6	\$1.36-\$1.59
Aluminum	572	503	331	72	0.102	8,90.6	\$1.23-\$1.4



7079-T6							1
Ti-6Al-4V	900	880	550	104	0.16	3,020	\$13.61-\$18.14
Ti-6AL-6V-2 Sn	1,050	980	760	112	0.164	3,000	\$14.53
T800S Carbon Fiber (Epoxy Composite)	3,290	N/A	N/A	294	0.065	6,686.6 (sublimes)	\$23.5

Originally, we picked aluminum 7075-T6 as the main material for our craft, which is a common choice. Aluminum 7075-T6 has widespread usage, with its strengths lying in its relative toughness and high resistance to fatigue. Its ultimate tensile strength is 572 MPa (82,961.6 psi), has a melting point of 890.6 °F, and only costs ~\$1.36 - \$1.59 per pound, making it a fantastic choice as the main material. However, after reevaluation and further research, we judged that using a carbon composite as the main material would be the best option. Carbon fiber is quickly becoming the industry standard, and many military aircraft use it in some capacity, such as the 5th-generation fighters like the F-35 and F-22. It offers a multitude of advantages over using aluminum, mainly down to its overall weight savings and relative strength. Compared to the aluminum alloy, T800S Carbon Fiber has an ultimate tensile strength of 3290 MPa (477,174.2 psi), nearly 6 times stronger. It also has high fatigue and corrosion resistance, increasing its durability, which is a key. It is not without its drawbacks, such as its issues with concentrated loads or delamination, but these issues can be mitigated by using a filament-reinforced composite like T800S and using correct layering to prevent shear loads from delaminating the material.

The fuselage, wings, and inlets will be shrouded in carbon fiber, using it mainly as a skin. Many aircraft use carbon fiber in this way, such as commercial craft like the Boeing 787 and military craft like the F-35. Carbon fiber has an extremely high load-bearing capacity and is resistant to cracks and general wear and tear. Furthermore, the RFP only calls for 2,000 hours of service life, which is well within the bounds of carbon fiber becoming worn. This will also reduce the increased costs associated with using composites, and structures can be offloaded to other materials, specifically metallic alloys. Therefore, we have decided to use Aluminum 7075-T6 alloy for our internal structures, in the fuselage and wing. In addition, for the landing gear, we decided to use the alloy Ti-6Al-4V due to



its relative lightweight, strength, and high corrosion and fatigue resistance. Lastly, it will also be utilized for engine mounting structure due to its strength and high melting point, in addition to Aluminum 7075-T6, as previously outlined in section 11.

10.2 Aircraft Load Analysis

The structural integrity and performance of an airplane wing are paramount to the overall safety and operational efficiency of an aircraft. The wing is subjected to significant aerodynamic and structural loads, often resulting in the concentration of maximum stresses within its components. To evaluate and enhance the structural performance of the aircraft, this report presents the results of a finite element analysis (FEA) conducted on a detailed 3D model of the left wing. The analysis is designed to provide comprehensive insights into the wing's structural behavior, accurately identify critical stress regions, and support informed recommendations for potential design optimizations to improve both safety and performance.

Due to the absence of direct pressure data, the finite element analysis was conducted using a calculated total wing load derived from the aircraft's maximum acceleration of 7g, corresponding to a force magnitude of 81,000 lb ($7 \times$ aircraft weight). The load distribution across the wing structure was approximated using the Schrenck distribution method, a standardized approach for estimating aerodynamic and inertial load gradients in symmetric aircraft configurations. Leveraging the geometric and load symmetry of the wing system, the analysis focused on simulating the left wing structure to reduce computational complexity.

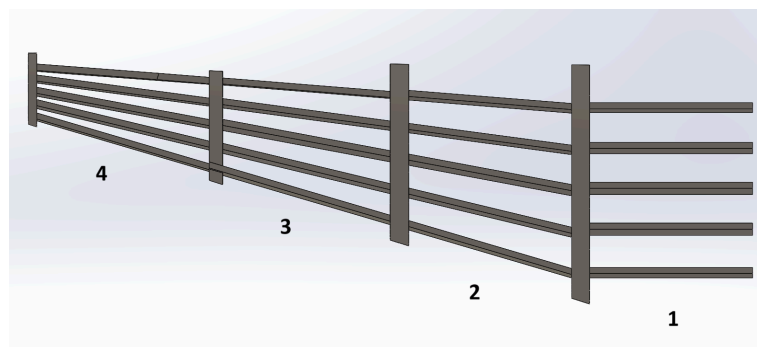


Figure 10.3.1: Subdivision of the Wing

The wing was divided into four distinct regions, with each section assigned an averaged load value based on the Schrenck distribution profile illustrated in Figure 10.3.1. Specific load magnitudes applied to these regions



are tabulated in Table 10.3.2. This segmentation ensured a representative approximation of stress gradients while maintaining computational efficiency.

Table 10.3.1: Specific Load Magnitudes on Wing

Section	Per Unit Length Load (lb/ft)
1	944.72
2	2834.18
3	4723.64
4	6613.10

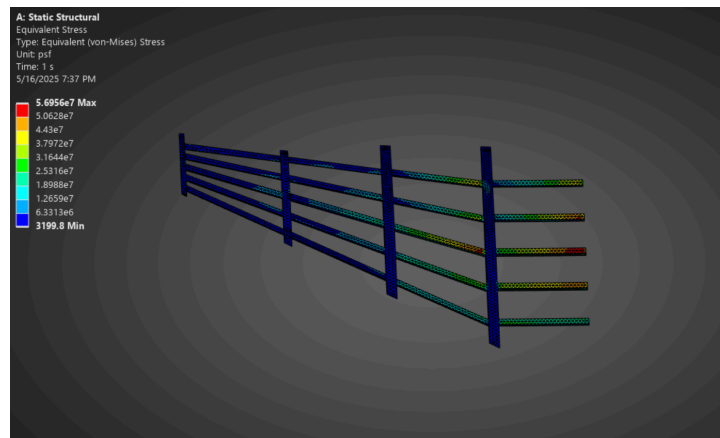


Figure 10.3.2: Maximum Stress Scenario for the Wing

The results indicate that the maximum von Mises stress of 57 MPa occurs at the wing root, the critical interface connecting the wing to the fuselage. This value remains well below the yield strength of the aluminum alloy frame (503 MPa), even when accounting for a safety factor of 1.5. The calculated allowable stress (503 MPa / 1.5 = 335 MPa) exceeds the observed maximum stress by a margin of 82%, confirming the structure's compliance with established safety standards under the specified loading conditions.

10.3 Mass Properties



As aforementioned, one of our main goals was to be as lightweight as possible. Our overall Max Design Gross Weight (DGW) was determined to be 12,000 lbs, based on the approximate weight of the T-38 and other lightweight fighter aircraft. However, this is easier said than done, as multiple internal components are not present in a craft like the T-38, and our craft also features a much larger engine that is overall lighter than the 2 J85-5A engines of the T-38. In addition, we have to consider the added AIM-120 AMRAAM & M61A1. With all of these factors in consideration, we have to make sure that our craft can achieve the 3 required missions, all while carrying payload such as fuel and weapons. Moreover, all of these components will affect our Center of Gravity (CG), so we have to ensure that these components are adequately spaced. To track, calculate, and adjust all of these figures, we utilized equations programmed in an .xlsx file. Our overall weights, findings, and calculations are shown below in Tables 10.4.1-10.4.8.

Table 10.4.1: Group Weight Statement

Group Weight Statement	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W ₀		Description
Max Design Gross Weight (DGW)	12000.00	36.00	0.00	1.037		Maximum Allowable Gross Weight
C.G. (% Cmac) at Max DGW		74.1%				
Gross Weight (w/margin)	11574.28	23.3	1.43	1.000		Bottoms Up Weight Estimate
C.G. (% Cmac) at Gross Weight		17.5%				
USEFUL LOAD ITEMS	5387.59	18.23	3.09	0.465	1.000	Sum of Crew, Cargo, Payloads, Fuel

Table 10.4.2: Payload

<i>Retained Payload</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W ₀	W/W _{UL}
External Store Suspension and Release Equipment	470.50	24.69	0.00	0.041	0.087
External pylon station 1	117.63	24.94	7.64	0.010	0.022
External Pylon 1	30.60	24.94	7.64	0.003	0.006
LAU-128 Launcher	87.03	24.94	7.64	0.008	0.016
External pylon station 2	117.63	24.44	5.96	0.010	0.022
External Pylon 2	30.60	24.44	5.96	0.003	0.006
LAU-128 Launcher	87.03	24.44	5.96	0.008	0.016
External pylon station 3	117.63	24.44	-5.96	0.010	0.022
External Pylon 3	30.60	24.44	-5.96	0.003	0.006



LAU-128 Launcher	87.03	24.44	-5.96	0.008	0.016
External pylon station 4	117.63	24.94	-7.64	0.010	0.022
External Pylon 4	30.60	24.94	-7.64	0.003	0.006
LAU-128 Launcher	87.03	24.94	-7.64	0.008	0.016
Gun	762.50	8.36	0.39	0.066	0.142
Gun - Uninstalled	275.00	10.60	0.65	0.024	0.051
Ammunition Feed System - Uninstalled	300.00	9.59	0.34	0.026	0.056
Ammunition Returned Casings	130.00	9.59	0.34	0.00	0.00
Gun Installation	57.50	10.20	0.00	0.005	0.011

<i>Expendable Payload</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W₀	W/W_{UL}
Internal Weapons	160.00	9.50	0.00	0.014	0.030
Internal Weapon 1	0.00	0.00	0.00	0.000	0.000
Internal Weapon 2	0.00	0.00	0.00	0.000	0.000
Gun Ammunition	160.00	9.50	0.00	0.014	0.030
External Weapons	1308.00	24.69	0.00	0.114	0.243
External Weapon 1	327.00	24.94	7.64	0.028	0.061
External Weapon 2	327.00	24.44	5.96	0.028	0.061
External Weapon 3	327.00	24.44	-5.96	0.028	0.061
External Weapon 4	327.00	24.94	-7.64	0.028	0.061

Table 10.4.3: Fuel

<i>Fuel</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W₀	W/W_{UL}
Internal Fuel - Usable	2534.52	23.85	0.00	0.219	0.470
Fuel - Trapped	152.07	23.85	0.00	0.013	0.028

Table 10.4.4: Contingency/Margin & Ballast

<i>Contingency/Margin and Ballast</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W₀	W/W_{UL}
EW Status Margin	562.43	36.00	0.00	0.050	0.091
EW Variation (undefined items)	0.00	0.00	0.00	0.000	0.000
Ballast	0.00	0.00	0.00	0.000	0.000
Design Margin Fraction	0.10				
Calculated Design Margin (lbm)	562.43				
EW Margin Used/Allocated	0.00				

Table 10.4.5: Propulsion Subsystem



<i>Propulsion Subsystem</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W₀	W/W_{UL}
Engine - Uninstalled	1360.00	40.06	0.00	0.118	0.220
Accessory Drive	81.60	40.06	0.00	0.007	0.013
Exhaust System	0.00	0.00	0.00	0.000	0.000
Engine Cooling	19.04	40.06	0.00	0.002	0.003
Oil Cooling	13.60	40.06	0.00	0.001	0.002
Engine Controls	30.60	40.06	0.00	0.003	0.005
Starter	21.76	40.06	0.00	0.002	0.004
Fuel System (includes Internal Tanks)	200.00	40.06	0.00	0.017	0.032
Other Propulsion Installation	136.00	40.06	0.00	0.012	0.022

Table 10.4.6: Additional Subsystems

<i>Subsystems</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W₀	W/W_{UL}
Vehicle Management System (VMS)	62.50	14.38	-0.98	0.005	0.010
VMS - Uninstalled	50.00	14.38	-0.98	0.004	0.008
VMS Installation	12.50	14.38	-0.98	0.001	0.002
Auxiliary Power Unit (APU)	125.00	20.09	-0.31	0.011	0.020
APU - Uninstalled	100.00	20.09	-0.31	0.009	0.016
APU Installation	25.00	20.09	-0.31	0.002	0.004
Hydraulics	85.88	0.00	0.00	0.007	0.014
Hydraulics - Uninstalled	68.70	0.00	0.00	0.006	0.011
Hydraulics Installation	17.18	0.00	0.00	0.001	0.003
Pneumatics	23.31	0.00	0.00	0.002	0.004
Pneumatics - Uninstalled	18.65	0.00	0.00	0.002	0.003
Pneumatics Installation	4.66	0.00	0.00	0.000	0.001
Electrical	275.00	8.30	0.01	0.024	0.044
Electrical - Uninstalled	220.00	8.30	0.01	0.019	0.036
Electrical Installation	55.00	8.30	0.01	0.005	0.009
Avionics	600.00	7.70	0.04	0.052	0.097
Integrated Comms, Nav, ID (ICNIA) - Uninstalled	100.00	17.89	0.45	0.009	0.016
Data Bus - Uninstalled	10.00	13.31	-0.97	0.001	0.002
Integrated Electronic Warfare System (INEWS) - Uninstalled	100.00	15.41	-0.12	0.009	0.016
IR Search and Track System (IRSTS) - Uninstalled	50.00	5.51	0.00	0.004	0.008
APG-81 Radar	220.00	4.00	0.00	0.019	0.036
Avionics Installation	120.00	0.00	0.00	0.010	0.019
Air Conditioning (ECS)	100.00	17.89	0.45	0.009	0.016



Air Conditioning - Uninstalled	10.00	13.31	-0.97	0.001	0.002
Air Conditioning Installation	100.00	15.41	-0.12	0.009	0.016

Table 10.4.7: Weights of Gear & Airframe Assemblies

<i>Alighting Gear</i>	Mass (lbm)	X-CG FS (ft)	Y-CG BL (ft)	W/W ₀	W/W _{UL}
Nose Gear Assembly	103.92	9.00	0.00	0.009	0.017
Nose Gear	103.92	9.00	0.00	0.009	0.017
Wheels and Tires	0.00	0.00	0.00	0.000	0.000
Main Gear Assembly	376.43	25.76	0.00	0.033	0.061
Main Gear	376.43	25.76	0.00	0.033	0.061
Wheels and Tires	0.00	25.76	0.00	0.000	0.000
<i>Airframe</i>	2026.64	27.49	0.00	0.175	0.328
Fuselage Assembly	1066.58	22.31	0.00	0.092	0.148
Fuselage	915.14	22.00	0.00	0.079	0.148
Air Induction System	141.43	24.17	0.00	0.012	0.023
Doors and Removable Panels	10.00	24.00	0.00	0.001	0.002
Exposed Wing Assembly	454.76	23.70	0.00	0.041	0.076
Exposed Wing	433.10	23.70	0.00	0.037	0.070
Installation and fittings	21.66	23.70	0.00	0.002	0.004
Horizontal Tail Assembly	196.92	40.83	0.00	0.018	0.033
Horizontal tail	187.54	40.83	0.00	0.016	0.030
Installation and fittings	9.38	40.83	0.00	0.001	0.002
Vertical Tail Assembly	308.38	42.50	0.00	0.028	0.051
Vertical Tail	293.70	42.50	0.00	0.025	0.047
Installation and fittings	14.68	42.50	0.00	0.001	0.002

Table 10.4.8: Configuration Characteristics, Performance Parameters, & Airframe Unit Weights

Configuration Characteristics	Value	
Sref (sq ft)	170.00	Wing Reference Area
Cmac (ft)	22.41	Used for CG Calculation
LECmac (ft)	19.39	Used for CG Calculation
Sht/Sref	0.25	Initial Horizontal Tail Sizing
Svt/Sref	0.13	Initial Vertical Tail Sizing
Mil Thrust (lbf)	6280	Engine Dry Thrust
Max Thrust (lbf)	9500	Engine Afterburner Thrust
Performance Parameters	Max DGW	Mid-Mission



Gross Weight	11574	10307.02
Wing Loading (psf)	68.08	60.63
T/W (Mil Power)	0.543	0.609
T/W (Max Power)	0.821	0.922
Airframe Unit Weights	UW (psf)	Sref (sq ft)
Fuselage	6.274	170.00
Exposed Wing	2.675	170.00
Horizontal Tail	4.633	42.5
Vertical Tail	13.954	22.1

Here, we include the expected takeoff weights for each of the 3 missions. Some assumptions are made, specifically for reserve and total fuel. A breakdown of the takeoff weights for each mission is given in Figure #. Overall, our total weight was found to be 10988.16 lbs, over 1,000 lbs under our DGW. This allowed us much room to operate and adjust our design as we were significantly underweight, giving us leeway with things such as materials and fuel. We calculated our reserve fuel to be 152.07 lbs, and the table shows how our craft will perform for each mission.

Using the equations from our mass properties sheet, we can successfully determine the weight of each major component of the craft. Some components, specifically the internal components, have been given set weights and volumes in the RFP and are simply added. These components include the avionics (ICNIA, Data bus, INEWS) Vehicle Management System, Fire Control Systems (IRSTS, Active Array Radar), APU, and the Air-to-Air Weapons (AIM - 120 AMRAAM, M61A1). Otherwise, general aerospace calculations, derived from Raymer's, are used to calculate component weights, such as the fuselage, landing gear, and wings [10]. After listing these weights, we added any necessary factors to take into account for weights or installations, such as an 85% multiplier for components that utilize composites, and a 5% multiplier for installation and fittings. Also listed are the exact locations of each component on our craft. After that, the sheet adds up all the associated weights, denoted by its Fuselage station (x-axis) and its Buttock Line (y-axis), both listed in feet. This allows us to determine the location of our CG as it is affected by the location and weight of each component. It can also be updated and adjusted in real time in case the CG is not in the correct position. With the total length of the craft being 40 ft, we got a CG of 23.69 ft. On its own, this number isn't completely useful, but we can compare it with our Center of Lift (COL) to determine the aerodynamic tendencies of our aircraft. We want our aircraft to be statically stable because of the RFP



requirement of an unaugmented subsonic longitudinal that is no greater than 10% and no less than -10%, so the CG needs to be positioned ahead of the COL. As aforementioned in Section 10, our COL was determined to be approximately 10 feet.

11. Repair, Maintenance, and Cost Analysis

11.1 Production Cases

Per the RFP, cost analysis for TITAN was conducted for production runs of 100, 500, and 1000 aircraft. The target flyaway cost per aircraft for a production run of 1000 aircraft was below \$25 million in 2024 USD.

11.2 RDT&E and Flyaway Cost Analysis

The Development and Procurement Costs of Aircraft (DAPCA-IV) model, developed by the RAND Corporation as outlined by Raymer, was used to estimate the RDT&E and flyaway costs [10]. The model estimates the number of hours for research, development, testing, and evaluation, and determines the cost by multiplying these hours by their respective wrap rates. The model also directly estimates the costs of materials manufacturing and engine production. It's pertinent to note that the DACPA-IV model estimates these costs in 2012 USD; consequently, an inflation rate was implemented to convert cost estimates into 2024 USD. This inflation rate was determined to be 37% based on United States CPI data provided by the US Bureau of Labor Statistics [19]. A fudge factor of 1.2 was then applied to all costs per Raymer's recommendation for modern military aircraft designs. It was assumed that 4 test aircraft would be used during flight testing. Table 11.2.1 highlights the total and unit RDT&E and flyaway costs for the production of 1000 aircraft. Additionally, Figure 11.2.1 shows the breakdown of the production costs for TITAN.

Table 11.2.1: RDT&E and Flyaway Cost Breakdown for 1000 Aircraft (2024 USD)

Category	Value
Engineering	\$1.252 billion
Tooling	\$795 million
Manufacturing	\$4.038 billion
Quality Control	\$537 million
Development Support	\$313 million



Flight Testing	\$115 million
Manufacturing Materials	\$2.09 billion
Engine Production	\$7.617 billion
TOTAL RDT&E AND FLYAWAY COST	\$16.758 billion
RDT&E AND FLYAWAY COST PER UNIT	\$16.758 million

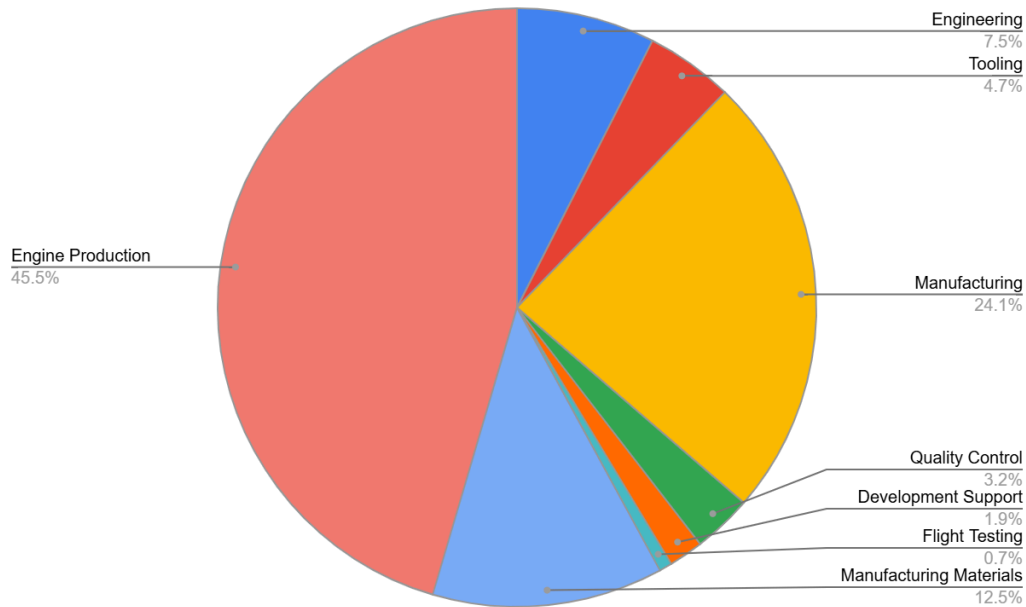


Figure 11.2.1: Production Cost Breakdown

The unit cost for each of the production runs is shown in Table 11.2.2 below. It should be noted that the avionics cost was excluded from cost calculations due to all avionic equipment being part of the GFE as outlined in the RFP.

Table 11.2.2: Unit Cost for Each Production Run (2024 USD)

Number of Aircraft Produced	Unit Cost
100	\$38.620 million
500	\$20.306 million
1000	\$16.758 million



11.3 Operations Cost

Using the DACPA-IV model, the total operations costs per production run were determined by taking fuel consumption and maintenance into consideration. For this analysis, the average yearly flight hours for TITAN were estimated to be 240 hours since this is the average yearly flight hours of current US military interceptors during peacetime [6]. The cost of JP-8 fuel is listed at \$3.52 per pound [12]. The ratio of maintenance man-hours to flight hours (MMH/FH) was determined to be 12.5 as the average of the range of Raymer’s approximations of MMH/FH for military fighters. The current hourly rate of an aircraft mechanic in the United States military is approximately \$36.54/hr [11]. Table 11.3.1 shows the total operations cost per year per aircraft.

Table 11.3.1: Operations Cost Per Year Per Aircraft (2024 USD)

Category	Value
Fuel	\$684,000
Maintenance	\$112,000
OPERATIONS COST PER UNIT	\$796,000

11.4 Life Cycle Cost

The life cycle costs were estimated at each production run used in section 15.2 by taking the sum of RDT&E, flyaway, and operations costs for each production run, the results of which are shown in Table 11.4.1 below.

Table 11.4.1: Lift Cycle Costs at Each Production Run (2024 USD)

Number of Aircraft Produced	Total Life Cycle Cost	Life Cycle Cost Per Unit
100	\$4.525 billion	\$45.253 million
500	\$13.470 billion	\$26.939 million
1000	\$23.391 billion	\$23.391 million

Based on Table 11.4.1, the life cycle cost per aircraft for a production run of 1000 TITAN aircraft is substantially cheaper than that of its modern-day American counterparts. Specifically, the life cycle of a TITAN aircraft comes out to be 8 years and 4 months from dividing the design service life of 2000 hours by its average



yearly flight hours. Over the same period, the cost to operate and sustain an individual F-35 is \$55 million in 2024 USD [9], over twice as expensive. Similarly, the life cycle cost of the F-22 Raptor is approximately \$198.4 million in 2024 USD over the same time [5], over 8 times as expensive. Consequently, the comparatively low life-cycle cost of TITAN makes it an affordable alternative to current American military interceptors.

11.5 Maintenance Approach

Maintenance for the HDI is a priority due to the emphasis on low-cost efficiency for the design and implementation of the aircraft. Having an appropriate maintenance plan will ensure an extended lifecycle for the fleet of aircraft for these missions. For TITAN, inspiration is taken from the Northrop T-38 Talon aircraft, including the maintenance process [13]. The T-38 Talon has its own TRIM program [14], which ensures serviceability for these aircraft for all training purposes. Additionally, this program allows the Air Force to account for the need for safe and reliable aircraft in a large fleet [16]. As for the TITAN, a similar program can be implemented to mass produce and service aircraft, following the RFP guidelines in a cost-efficient manner. Mentioned in earlier sections, the TITAN is an affordable and cheaper aircraft compared to most other American counterparts. Keeping this cost low will mostly rely on how well servicing can take place.

A few key philosophies can be implemented for the maintenance approach for TITAN. The first being Single Layer Direct Access for all components. This allows for all components to be accessible without the removal of other components in the process. Appropriate distribution of internal components and maintenance ports will accommodate this important aspect. Additionally, keeping some components together or in a close vicinity will not only add to the aforementioned direct access approach, but it will also ensure fewer sources to check to locate problems. To also follow proper standards, technical orders and work cards should be created for any servicing. To create these documents and other relevant documents [1], military standards should be adhered to. For example, following the military standards of MIL-STD-38784 (Formatting) and MIL-HDBK-516C (Airworthiness) will keep all plans for maintenance in a consistent manner that will keep these TITAN aircraft in the air as long as possible and with minimal servicing periods.

11.5.1 Maintenance Ports



Choosing proper maintenance port locations along the fuselage will allow for the aforementioned maintenance approach to be feasible. Based on the internal layout from space constraints and standard aircraft layouts, locations for the ports can be determined. Figure 11.5.1 shows the frontmost access point, which is simply the nose cone that opens by a hinge mechanism.

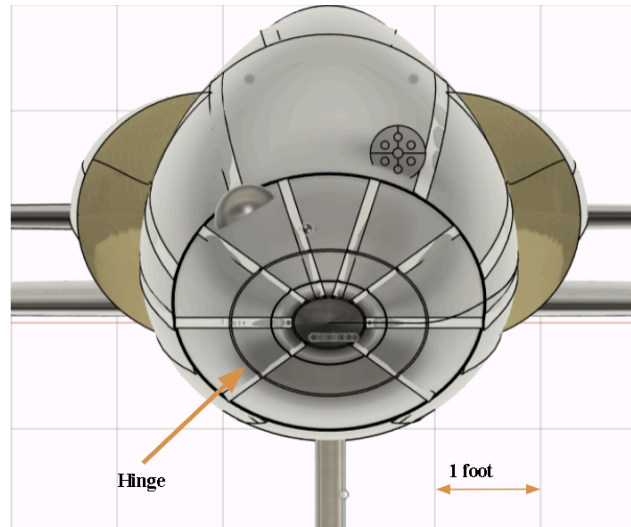


Figure 11.5.1: TITAN Front View

The radar and tracking components are both housed in the nose section and have single-layer direct access through the nose cone. Typically, these components are housed towards the front of the fuselage to prevent excessive interference with other components. Additionally, for this design, these components are placed ahead of the M61A1 cannon, which will prevent barrel gases from interfering with the tracking system. Following this, the next set of internal components is located in the front-middle fuselage, which lies behind the radar but in front of the wings. Figure 11.5.2 shows the maintenance approach for this section of the aircraft.



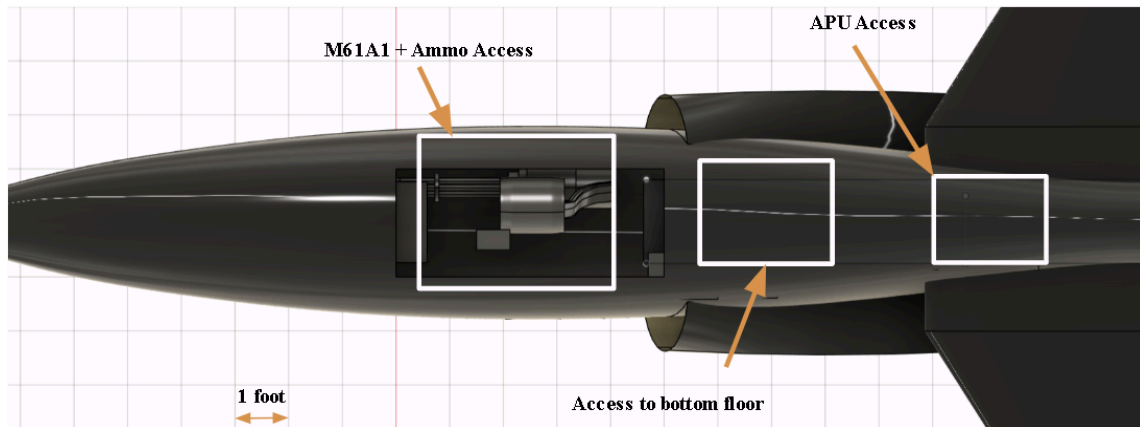


Figure 11.5.2: TITAN Bottom View - Maintenance Ports

A bottom view shows access to the remaining internal components and the APU through two different ports. Additionally, access to the M61A1 and ammo drum is vital, specifically from underneath the fuselage, which allows for easier accessibility. While there is not a separate port for this weapon, the front landing gear bay, while open, has ample space to access the gun in its mounted position, along with the ammo drum. Once again, all components here have single-layer direct access, especially the internal components listed in the “bottom floor” region, which is simply a plate that separates components above and below it.

Next, a bottom view of one of the wings is shown in Figure 11.5.3. In this view, the landing gear and bay are hidden for easier viewing, although the white rectangle towards the bottom represents the area where the bay is located. When extended, the bay can be used to access internal wiring and service any issues.



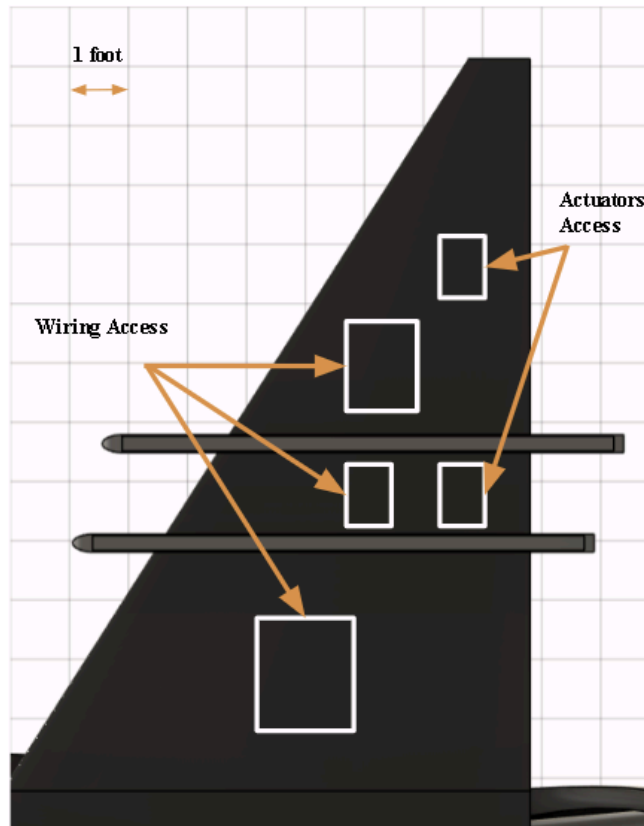


Figure 11.5.3: TITAN Wing View - Maintenance Ports

With this view, port locations are distributed for multiple purposes. As mentioned above, internal wiring can be accessed both through the landing gear bay and along the wing surface, from below. A middle port is included to access wires and parts of the rail launcher for the AMRAAMs, in case there are any issues that arise within the wing layout. Additionally, actuator ports are incorporated for easy servicing to ensure proper control surface operation.

Finally, Figure 11.5.4 shows a rear fuselage view in regards to engine servicing. The engine is the main component in this part of the aircraft, and maintenance is important to ensure the operation of TITAN.



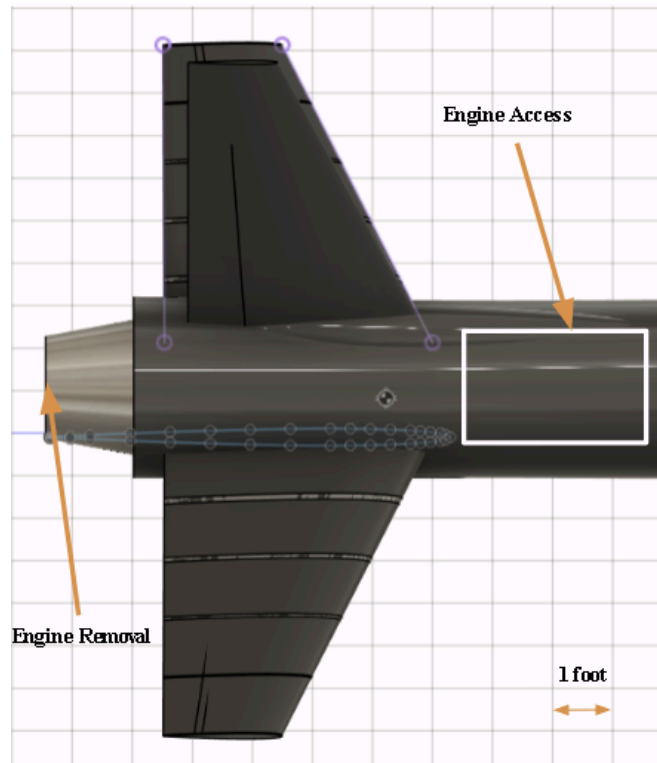


Figure 11.5.4: TITAN Rear Fuselage View - Maintenance Ports

A maintenance port can be placed on the side of the fuselage to access the engine without full removal. In case the engine needs to be removed for additional inspection, a similar approach to the T-38 Talon aircraft can be taken, and the engine can simply be removed from the back.



12. Conclusion

The TITAN Homeland Defense Interceptor represents a well-balanced response to the RFP's requirements, achieving superior performance, cost efficiency, and mission adaptability through disciplined design decisions and rigorous engineering analysis. Designed around the principles of simplicity, survivability, and affordability, the aircraft fulfills its three core mission profiles: Defensive Counter-Air, Point Defense Interception, and Intercept/Escort.

By leveraging a proven airframe lineage, integrating a non-developmental F125-IN engine, and prioritizing internal system efficiency, the TITAN meets or exceeds all performance and cost thresholds outlined in the RFP. The aircraft delivers the required supersonic capability, endurance, and weapons capacity while remaining under the \$25 million unit flyaway cost cap. Its compact, aerodynamically efficient airframe, internal cooling and fuel systems, and remote-piloted capability ensure mission success in well-supported environments.

Through the integration of lessons learned from legacy systems and modern innovations in materials, avionics, and propulsion, TITAN offers a viable and sustainable solution for homeland air defense. The result is an aircraft that is not only effective in combat but also optimized for maintainability, lifecycle affordability, and rapid deployment. With scalable production costs, robust mission performance, and support for remote pilot operation, TITAN stands ready to serve as the cornerstone of a modern, responsive air defense fleet.



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